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Abbreviations

AO Aero-Optical

APL Applied Physics Laboratory

BL Boundary Layer

BO Born-Oppenheimer

BSE Bore-Sight Error

CFD Computational Fluid Dynamics

CGS Centimeter-Gram-Second units

CSBA Center for Strategic and Budgetary Assessments

DES Detached Eddy Simulation

DNS Direct Numerical Simulation

DoD Department of Defense

DoE Department of Energy

dof degrees of freedom

E&M Electromagnetic

FDM Finite Difference Method

FEM Finite Element Method

FVM Finite Volume Method

G-D Gladstone-Dale

HAOT Hypersonics Aerodynamics Optics Tool

HPC High Performance Computing

HyNECC Hypersonic NonEquilibrium Comparison Cases

JHU Johns Hopkins University

LANL Los Alamos National Laboratory

LeMANS Le Michigan Aerothermodynamic Navier-Stokes Solver

LES Large Eddy Simulations

LLNL Lawrence Livermore National Laboratory

Mutation++ MUlticomponent Thermodynamic And Transport properties for IONized gases in C++

NASA National Aeronautics and Space Administration

NEMO NonEquilibriumMOdeling

N-S Navier-Stokes

OPD Optical Path Difference

OPL Optical Path Length

PDE Partial Differential Equation

PyPI Python Package Index

QM Quantum-Mechanics

RANS Reynolds Averaged Navier-Stokes

SGS Sub-Grid Scales

SI International System of Units

SU2 Stanford University Unstructured

URANS Unsteady Reynolds Averaged Navier-Stokes

WD Wavefront Distortion

ZPE Zero Point Energy

Nomenclature

Upper-case Roman

B_e	Rotational constant in equilibrium position $\left[cm^{-1}\right]$
D_e	Centrifugal distortion constant $\left[cm^{-1}\right]$
$\mathcal{E}_{elect}$	Electronic energy
$\mathcal{E}_{kinetic}$	Kinetic energy
$\mathcal{E}_{potential}$	Potential energy
$\mathcal{E}_{rot}$	Rotational energy
$\mathcal{E}_{trans}$	Translational energy
$\mathcal{E}_{vib}$	Vibrational energy
$\mathcal{F}^c$	Convective fluxes
$\mathcal{F}^v$	Viscous fluxes
$\mathcal{H}$	Hamiltonian
J	Rotational quantum number
K_{GD}	Gladstone-Dale constant
M_s	Species molar mass
N_i	Species number density
$\mathcal{Q}$	Source terms
T	Temperature
$\mathcal{U}$	Conservative variables
V	Volume
Z	Partition Function

Lower-case Roman

c_m	Speed of light in a medium
c_p	Heat capacity at constant pressure
c_v	Heat capacity at constant volume
e	Internal energy
g_i	Degeneracy
h	Enthalpy
p	Pressure
q_e	Electron volt constant
n	Index of refraction
r_e	Equilibrium internuclear distance $\left[\mathring{A}\right]$
u^+	Normalized velocity
u_τ	Friction velocity
v	Vibrational quantum number
y^+	Normalized wall distance

Lower-case Greek

α	Polarizability $\left[m^3\right]$
α_e	Rotational constant, first term $\left[cm^{-1}\right]$
β	Thermodynamic beta $\left(\frac{1}{k_B T}\right)$
γ	Specific heat ratio
ϵ_m	Dielectric constant in a medium
μ_m	Magnetic constant in a medium
μ_w	Dynamic Viscosity at the wall
ρ_m	Density of the medium
ρ_w	Density at the wall
$\tau_{chemical}$	Chemical reaction characteristic time
τ_{flow}	Flow characteristic time
τ_{trans}	Translational mode characteristic time
τ_{rot}	Rotational mode characteristic time
τ_{vib}	Vibrational mode characteristic time

τ_w	Shear stress at the wall
ω	Signal frequency $\left[\frac{rads}{sec}\right]$
ω_e	Vibrational constant, first term $\left[cm^{-1}\right]$
$\omega_e x_e$	Vibrational constant, second term $\left[cm^{-1}\right]$
$\omega_e y_e$	Vibrational constant, third term $\left[cm^{-1}\right]$
ω_0	Ground-level angular frequency

Physical Constants

R	Gas constant, 8.3145 $\left[\frac{J}{mol\,K}\right]$
c_0	Speed of light in a vacuum, $3e8$ $\left[\frac{m}{s}\right]$
$\hbar$	Reduced Plack constant, $1.054e{-}34$ $[Js]$
k_B	Boltzmann constant, $1.380e{-}23$ $\left[\frac{J}{K}\right]$
ϵ_0	Dielectric constant in vacuum, $8.8854e{-}12$ $\left[\frac{C}{Vm}\right]$
μ_0	Magnetic constant in a vacuum, $4\pi e{-}7$ $\left[\frac{H}{m}\right]$

Chapter 1

Introduction

1.1 Motivation

Several reports in the classified and unclassified [1], [2] environment have stated the concern of aero-optic effects at hypersonic speeds to be of paramount importance for national security. Applied work in this field is not only being done by national laboratories such as Lawrence Livermore National Laboratory (LLNL), Johns Hopkins University (JHU)-Applied Physics Laboratory (APL), Los Alamos National Laboratory (LANL) but also by the Department of Defense (DoD) contractors such as Lockheed Martin Corp., Northrop Grumman Corp., Raytheon Technologies. As expected most of this work is limited in the open literature. However, work that investigates the effect of optical properties in a hypersonic environment is lacking and there are many fundamental phenomenons such as index of refraction, that needs to be better understood.

Fig. 1.1 illustrates the notional flight paths for various long-range weapons. Accurate target detection is of paramount importance in these scenarios. Cruise missiles generally travel at subsonic speeds, whereas ballistic missiles and boost-glide vehicles can reach speeds well above the speed of sound. At these hypersonic velocities, the vehicles are subject to intense aerodynamic heating and dynamic pressure, resulting in complex hypersonic flow fields around the vehicle. These conditions can significantly degrade seeker performance, impacting sensors' ability to lock onto targets due to thermal interference, signal distortion, and increased drag forces. Advanced guidance systems must compensate for these factors to maintain precision in target acquisition and engagement.

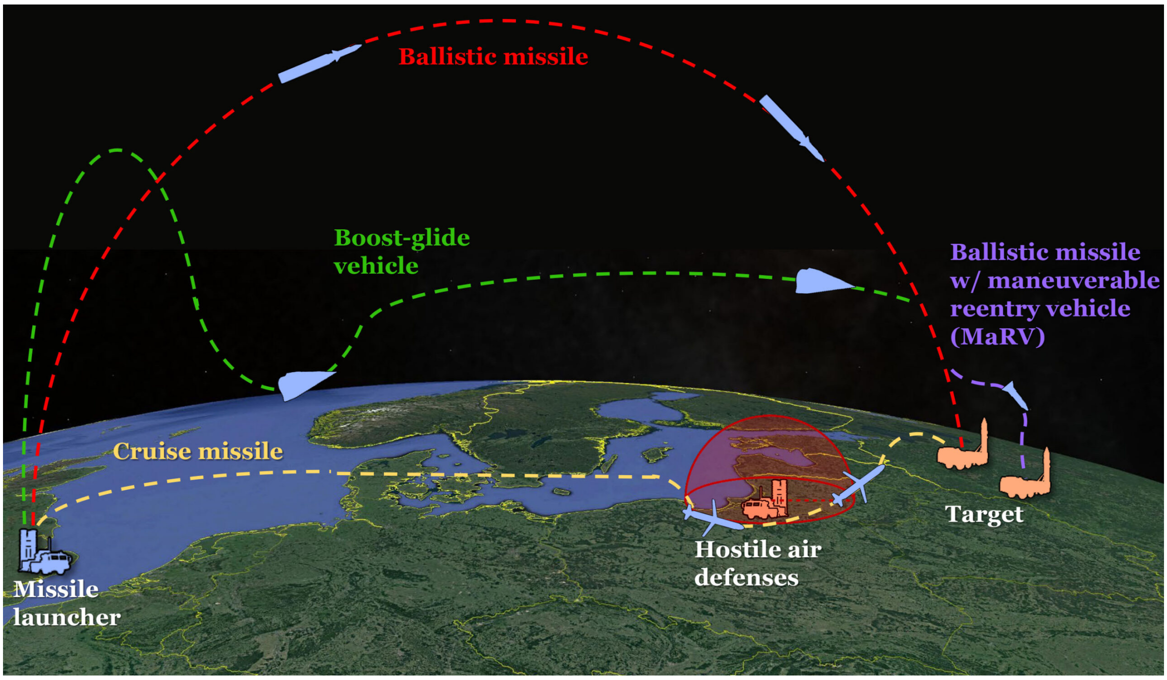


Figure 1.1: Notional flight paths of hypersonic boost-glide missiles, ballistic missiles, and cruise missiles. (CSBA graphic)

A crude approximation for the index of refraction in the atmosphere at radio frequencies was proposed by Smith and Weintraub [3] and it is given on eq. (1.1). Where $K_1 = 79 \left[\frac{K}{mB}\right]$, $K_2 = 4800$ $[K]$, e is the partial pressure of water vapor in millibars.

$$n(h) \approx 1 + \frac{K_1}{T(h)} \left(p(h) + K_2 \frac{e(h)}{T(h)} \right) \tag{1.1}$$

Fig. 1.2 illustrates the index of refraction as a function of altitude for a *blue-sky* scenario. The most drastic change in the index of refraction occurs at approximately 15 $[km]$ above sea level, which is in the tropopause, near the boundary where the troposphere ends and the stratosphere begins. This is also the altitude at which commercial planes and jets typically fly. It is a critical region where sensor accuracy is especially important.

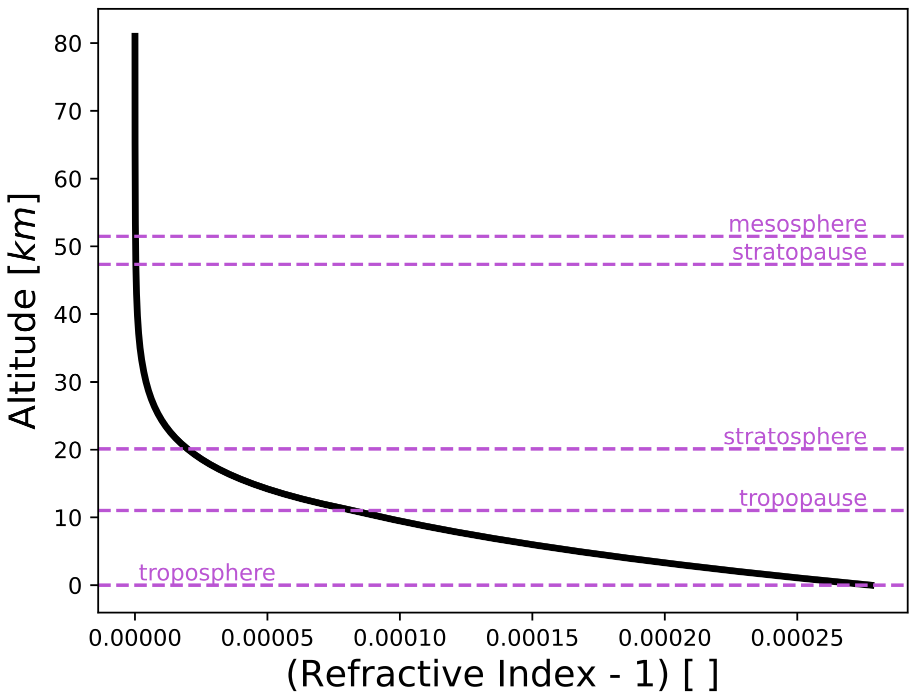


Figure 1.2: Atmospheric index of refraction

Understanding the effects of the environment in which an optical beam traverse is imperative in target detection. Radar engineers will be mainly concerned with the atmospheric environment (rain, clouds, fog, snow) but will not much much consideration on the impact aerodynamic effects such as shock wave and boundary layer interactions has on the optical beam. Some work showing how much the index of refraction is affected by aerodynamic effects has been performed by Gupta et.al. [4] for supersonic flows and showed that there is indeed, a change on index of refraction across a shock wave. For subsonic flows, it might be a valid assumption to ignore the aerodynamic effects on the optical beam; however for supersonic and specially for hypersonic flows this assumption has to be revisit. During a hypersonic flight the temperatures behind the shock are very high, that will no only induce thermal noise on the sensor but will also affect the electromagnetic properties that will influence the sensor's performance. Hence a higher distortion as well as power loss will be expected in a hypersonic flow. Being able to accurately understand this is of importance to build accurate sensor that can not only target objects moving at these speeds, but can also accurate predict targets from an object moving at hypersonic speeds.

Fig. 1.3, shows the path that a beam in a hypersonic flight will follow. The beam has to travel across the isolated gas, the radome, the boundary layer, shock layer and lastly environment. Traditionally, radar engineers ignore aerodynamic effects (shock waves, expansion waves) for their analysis; however for the case of hypersonic flights the aerodynamic effects cannot be ignored anymore, because the electromagnetic properties across the boundary layer and shock layer change.

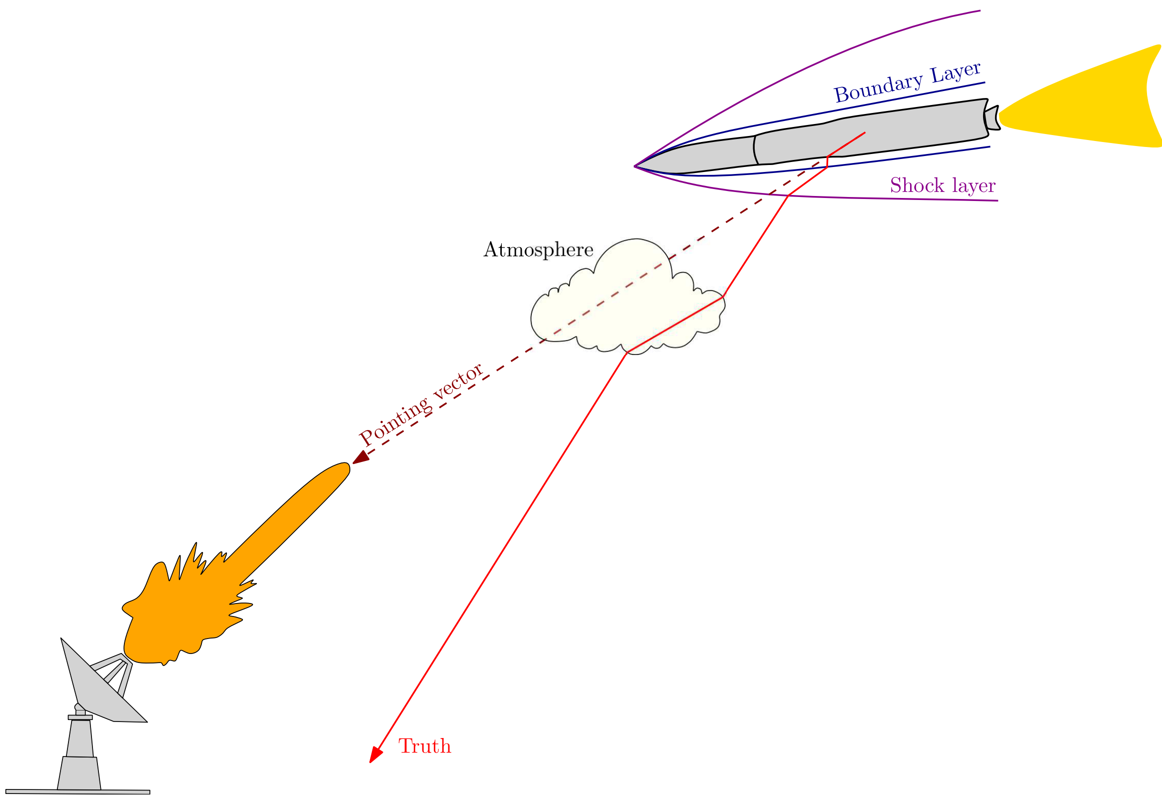


Figure 1.3: Optics Schematic

Aero-optical investigations can be split into four quadrants as shown in tab. 1.1. The optical parameters are contingent on the degree to which turbulence eddy resolution and turbulence fluctuations are captured in the Computational Fluid Dynamics (CFD) simulation. In cases where a lower level of flow resolution is utilized, either a laminar flow or a Reynolds Averaged Navier-Stokes (RANS) simulation are employed as shown in Quadrants I and III of tab. 1.1. While a RANS approach averages out turbulent fluctuations, it is a popular approach for hypersonic vehicle analysis due to it's efficiency [5] and can still provide aero-optical quantities such as G-D constant, polarizability, Bore-Sight Error (BSE), Optical Path Length (OPL), Optical Path Difference (OPD), and Wavefront Distortion (WD). The amount of enthalpy is also an important criteria in the flow field. With high enthalpy flows, many physical phenomena in the flow field that are not present in low-speed flows present themselves. This includes nonequilibrium, reactive flows, and excitation of molecular internal modes, all of which can affect optical signals either directly or indirectly; these are shown in the top Quadrants. This work aims to investigate and compare all quadrants using different fidelities of nonequilibrium and fluid flows.

Quadrant III: High-Enthalpy, Low-Flow Resolution *Physics captured:* chemical and thermal nonequilibrium, shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD.	**Quadrant IV:** High-Enthalpy, High-Flow Resolution *Physics captured:* chemical and thermal nonequilibrium, shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD, OPD.
Quadrant I: Low-Enthalpy, Low-Flow Resolution *Physics captured:* shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD.	**Quadrant II:** Low-Enthalpy, High-Flow Resolution *Physics captured:* shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD, OPD.

Table 1.1: Schematic of the fidelity of aero-optical simulations.

A conference paper focusing on quadrants I and III was published by Liza et al. [6]. In this paper, the authors examined the impact of high temperatures and nonequilibrium conditions on polarizability. Additionally, they investigated the effects of Mach number, turbulence, and chemistry modeling on optical properties, including the index of refraction and the dielectric constant of air.

1.2 Dissertation Outline

Missing dissertation outline

Chapter 2

Theory

This chapter discusses the theoretical foundations supporting this work. It provides essential background in hypersonic flows, optics, quantum mechanics, and turbulence, covering the principles and frameworks necessary to understand and contextualize the research.

2.1 Hypersonic Overview

The first hypersonic flight took place during the early stages of the Cold War, in 1949 when the German V2 rocket developed during the Second World War in Nazi Germany and brought to the USA (possible during operation Paperclip) reached hypersonic speeds as it entered the atmosphere [7]. In 1961, the Soviet Union cosmonaut Yuri Garin became the fist person to safely reach space and orbit Earth. During his journey he became the first human in history to travel at hypersonic speeds [7].

As the velocity of the flow increases, compressibility effects can not be ignored. The flow field is typically characterized in five distinct regimes: incompressible, subsonic, transonic, supersonic and hypersonic [8].

- **Incompressible:** A flow characterized by a constant density in which the density doesn't change significantly even when subjected to pressure variations.

- **Subsonic:** A flow characterized by a Mach number less than 1 at every point. In this regime, compressibility effects can generally be neglected. The flow patterns in a subsonic flow are typically smooth and continuous, with no shock wave formation. Due to the absence of shock waves, these flows are often considered isentropic.

- **Transonic:** A flow characterized by a Mach number ranging between 0.8 to 1.2. In this regime, regions of subsonic and supersonic coexist; creating complex flow patterns and leading to the formation of localized shock waves. Compressibility effects cannot be ignored, leading to variations in fluid density.

- **Supersonic:** A flow characterized by a Mach number greater than 1. In this regime compressibility effects cannot be ignored, leading to substantial variations in fluid density. Shock waves are present in this regime which produces discontinuous changes in pressure, temperature and density.

- **Hypersonic:** A flow characterized by a Mach number greater than 5. In this regime strong shock waves are present, leading to abrupt and significant changes in pressure, temperature, and density. Due to the high velocities, the kinetic energy in the flow is converted to thermal energy resulting in high temperatures. These high temperatures lead to *real gas* effects and a thicker boundary layer.

At higher velocities, particularly in the supersonic and hypersonic regimes, compressibility effects are crucial to understanding the flow dynamics, and the simple models used at lower speeds are no longer adequate. fig. 2.1 show these regions, as the Mach number increases, shock waves start to form and compressibility effects become more important.

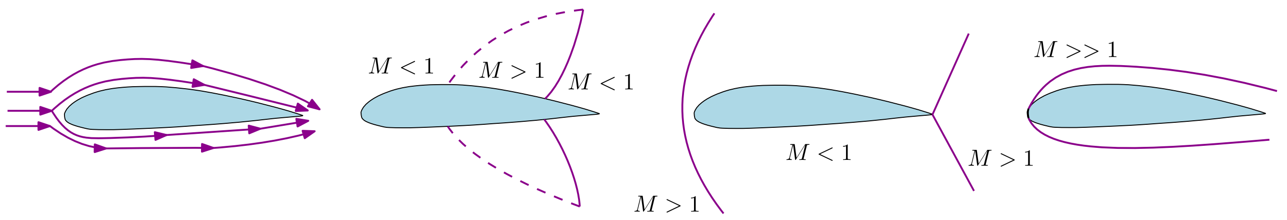


Figure 2.1: Mach number regions

In a hypersonic flow, the gas effects are typically called *real gas* effects, which can be misleading. Four categories of gases can be identified [9], [10].

- **Calorically Perfect Gas:** A gas with constant specific heats (equations (2.1a) and (2.1b)) in which the enthalpy and internal energy (equations (2.1c) and (2.1d)) are functions of temperature.

$$c_p = \text{constant} \tag{2.1a}$$

$$c_v = \text{constant} \tag{2.1b}$$

$$h = c_p T \tag{2.1c}$$

$$e = c_v T \tag{2.1d}$$

The *perfect gas* equation of state (eq. (2.2)) holds.

$$pv = RT \tag{2.2}$$

- **Thermally Perfect Gas:** A gas where specific heats (equations (2.3a) and (2.3b)) are functions of temperature only, in which the enthalpy and internal energy (equations (2.3c) and (2.3d)) are functions of temperature. The *perfect gas* equation of state (eq. (2.2)) holds.

$$c_p = c_p(T) \tag{2.3a}$$

$$c_v = c_v(T) \tag{2.3b}$$

$$h = h(T) \tag{2.3c}$$

$$e = e(T) \tag{2.3d}$$

- **Chemically Reacting Mixture of Perfect Gases:** A multispecies, chemically reacting gas in which intermolecular forces are neglected and each individual species obeys the *perfect gas* law eq. (2.2). The heat capacities (equations (2.4a) and (2.4b)) not only depend on temperature but also in the number of particles per chemical specie. The enthalpy and internal energy (equations (2.4c) and (2.4d) are functions of temperature and number of particles per chemical specie.

$$c_p = c_p(T, N_1, N_2, ..., N_n) \tag{2.4a}$$

$$c_v = c_v(T, N_1, N_2, ..., N_n) \tag{2.4b}$$

$$h = h(T, N_1, N_2, ..., N_n) \tag{2.4c}$$

$$e = e(T, N_1, N_2, ..., N_n) \tag{2.4d}$$

- **Real Gas:** A gas in which the effect of intermolecular forces is considered. Specific heats (equations (2.5a) and (2.5b)) are functions of temperature and pressure or volume). Due to intermolecular forces, the enthalpy and internal energy (equations (2.5c) and (2.5d) are functions of temperature and pressure or volume.

$$c_p = c_p(T, p) \tag{2.5a}$$

$$c_v = c_v(T, v) \tag{2.5b}$$

$$h = h(T, p) \tag{2.5c}$$

$$e = e(T, v) \tag{2.5d}$$

For this gas, the *perfect gas* equation eq. (2.2) is no longer valid and a real gas equation such as the *Van der Waals* eq. (2.6) can be used. a and b are constants that depends on the type of gas.

The *Van der Waals* model, makes two modifications to the ideal gas equation eq. (2.2), adding a term to the pressure and subtracting a term to the volume. The term modifying the pressure, accounts for the short-range attractive forces between molecules that are not touching (potential energy repulsion). The term modifying the volume ensures that the volume does not go down to infinity. The constant b represents the minimum volume occupied by a molecule when it is "touching" all its neighbors [11].

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT \tag{2.6}$$

In hypersonic flow, where speeds exceed Mach 5, the kinetic energy of the gas molecules is largely converted into thermal energy, resulting in extremely high temperatures. At these velocities, the chemical composition of the gas plays a critical role in the behavior of the flow. The high temperatures can cause significant molecular dissociation (the breaking apart of molecules) and ionization (the formation of charged particles) of air molecules. For example, at temperatures above 2000 $[K]$, molecular oxygen begins to dissociate, and it is fully dissociated at around 4000 $[K]$. Similarly, molecular nitrogen starts to dissociate above 4000 $[K]$, and complete dissociation occurs at temperatures exceeding 9000 $[K]$ [10].

At these elevated temperatures, the internal energy modes of gas molecules are significantly altered. The Hamiltonian of the system, which represents the total kinetic and potential energy, must be modified to account for additional degrees of freedoms (dofs). These additional energy modes include rotational, vibrational and electronic energies, which become accessible due to the high energy levels present in hypersonic flows. Eq. (2.7) shows how the additional energy modes in a hypersonic flow.

$$\mathcal{H} = \mathcal{E}_{kinetic} + \mathcal{E}_{potential} = \underbrace{\mathcal{E}_{trans} + \mathcal{E}_{rot} + \mathcal{E}_{vib} + \mathcal{E}_{elect}}_{\text{diatomic}} = \underbrace{\mathcal{E}_{trans} + \mathcal{E}_{elect}}_{\text{monatomic}} \tag{2.7}$$

This modification in the Hamiltonian is critical because the traditional energy models, which assume only translational motion, are insufficient to describe the thermodynamic behavior at such extreme conditions.

Fig. 2.2 illustrates the additional degrees of freedom that emerge in a hypersonic flow. As molecules absorb increasing amounts of energy, rotational and vibrational modes are excited, followed by electronic excitations and, at sufficiently high temperatures, ionization. These phenomena significantly affect the thermodynamic properties of the gas, including heat capacity, thermal conductivity, and viscosity, all of which must be accurately modeled to predict the behavior of hypersonic flows.

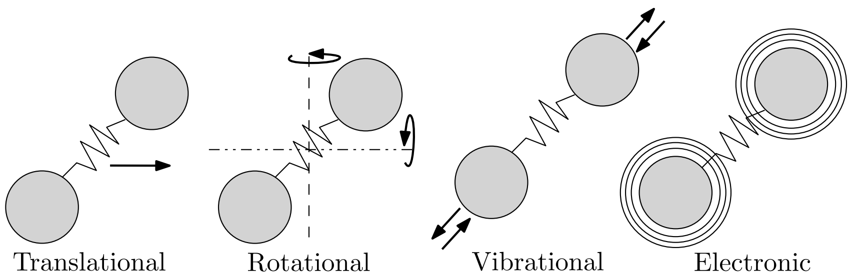


Figure 2.2: Additional Degrees of Freedom

In addition, each newly accessible energy mode operates on different time scales, which significantly influences the thermodynamic response of the gas in hypersonic flows. These time scales are associated with the energy required for the excitation of rotational, vibrational, and electronic modes, as well as ionization. For instance, rotational modes tend to equilibrate quickly, while vibrational and electronic excitations, along with dissociation and ionization, occur over longer time scales due to the higher energy thresholds involved.

Eq. (2.8) shows the specific time scales associated with each energy mode. Understanding these time scales is crucial for accurately modeling the non-equilibrium thermodynamic processes in hypersonic regimes.

$$\tau_{flow} \approx \tau_{trans} < \tau_{rot} < \tau_{vib} < \tau_{chem} \tag{2.8}$$

The different time scales is the reason why flows at hypersonic speeds are in nonequilibrium. After air molecules travel across a shock, it will take certain number of collisions to convert the energy into thermal energy. In the case of translational energy is almost instantaneous, in the case of rotational is approximately 10 collisions and in the case of vibrational is approximately 1000 collisions [12]. The nonequilibrium characteristics of the flow can be characterize by comparing the flow characteristic time with the other time scales [13].

These additional energy modes increase the dof of the system, allowing molecules to occupy more energy levels (rotational, vibrational, electronic). Due to this, the partition function (which indicates how particles in an assembly are partitioned among various energy levels of an atom or molecule [14]) given in eq. (2.9) has to be modified to consider these additional dofs. In statistical mechanics, the partition function is used to calculate the heat capacities of the system, which are then used to calculate macro properties such as heat transfer and temperature, which are employed in fluid dynamics.

$$Z = \sum_i g_i e^{-\beta E_i} \tag{2.9}$$

The thermodynamic beta is given by eq. (2.10)

$$\beta = \frac{1}{k_B T} \tag{2.10}$$

Fig. 2.3 shows the change in heat capacity when additional dofs are accessible.

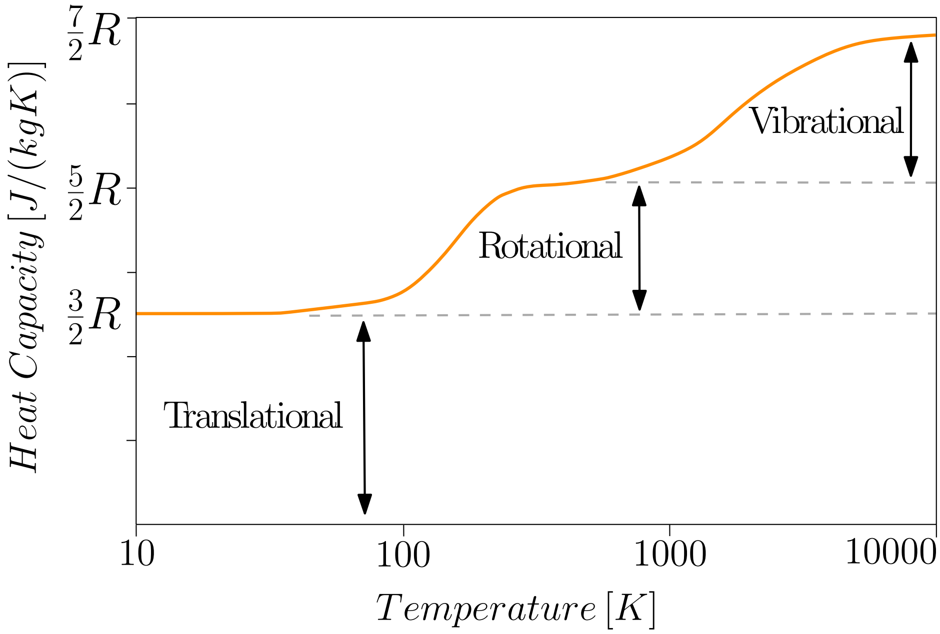


Figure 2.3: Heat Capacity

Under these conditions, the calorically perfect gas approximation becomes unsuitable, as it neglects key thermochemical effects that arise at high velocities fig. 2.4 illustrates the pre-shock and post-shock temperatures across various altitudes. As expected, at high speeds, the calorically perfect gas model underestimates the temperature increase across the shock.

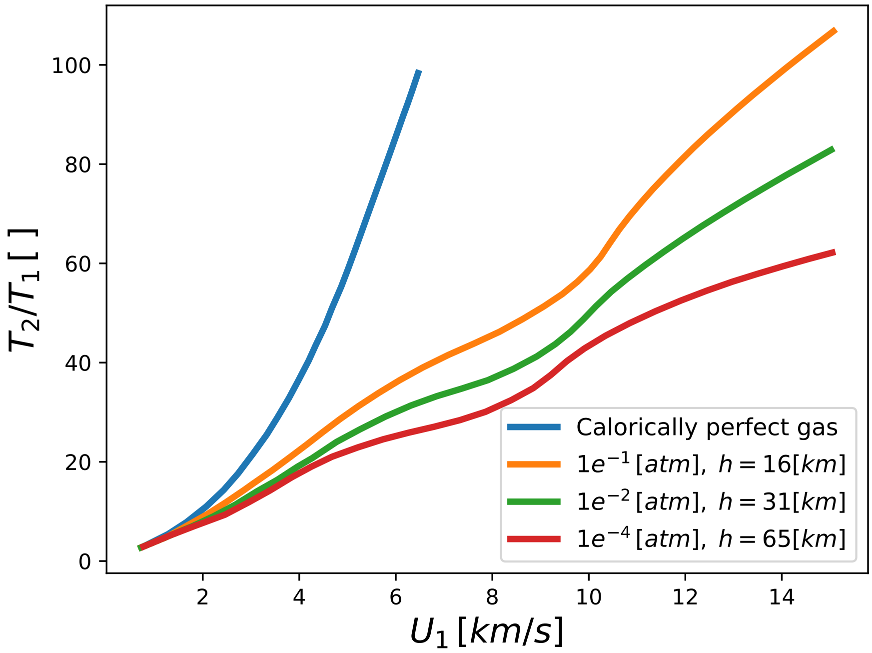


Figure 2.4: Temperature ratio across a shock wave

At elevated velocities, particularly in the hypersonic regime, the assumptions of constant specific heats and the absence of chemical reactions no longer hold. The gas molecules experience significant excitation, dissociation, and ionization, which are critical thermochemical effects that must be accounted for. These processes lead to non-linear increases in temperature and energy exchange across the shock, rendering the calorically perfect gas model insufficient [15].

The thermochemical effects encountered in hypersonic flows are directly influenced by the interplay between the gas's internal energy modes and the time scales of the flow. Whether the gas behaves in an equilibrium or frozen state depends on the relative rates of these energy exchanges and the flow dynamics. When the energy exchange rates are faster than the flow time scales, equilibrium can be achieved, allowing for uniform distribution of energy across all modes. However, when the flow dynamics outpace these energy exchanges, the gas is unable to equilibrate, leading to frozen behavior. This dynamic directly links the state of the gas to the complex thermochemical changes observed in hypersonic conditions, such as those caused by excitation, dissociation, and ionization. Fig. 2.5 shows the difference between an equilibrium and frozen flow after a nozzle for the temperature and mass fraction.

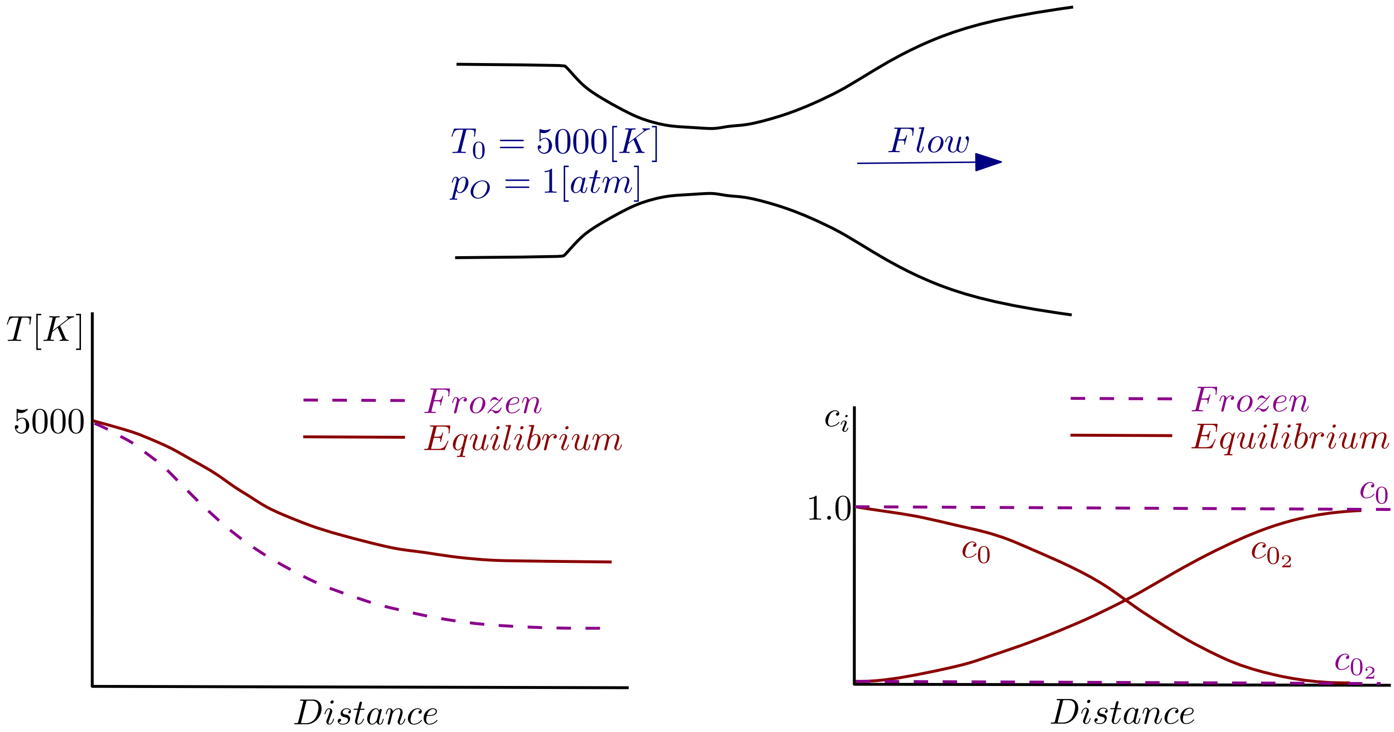


Figure 2.5: Frozen and Equilibrium plots across a nozzle

- **Equilibrium Flow:** A gas is said to be in equilibrium when it has sufficient time to allow all its internal energy modes—translational, rotational, vibrational, and possibly electronic—to exchange energy and reach a uniform state of equilibrium. In this condition, the distribution of energy within these modes aligns with the thermodynamic properties predicted by statistical mechanics, such as the Boltzmann distribution. Equilibrium assumes that the rate of energy exchange between modes is much faster than the flow dynamics, ensuring the gas behaves in a thermodynamically balanced state.

- **Frozen Flow:** Frozen flow occurs when a gas moves so rapidly that its internal energy modes (translational, rotational, vibrational, or chemical) do not have sufficient time to equilibrate during the flow. This happens because the flow time scale is much shorter than the time required for internal energy exchanges or chemical reactions to occur. In this state, the internal energy distribution remains effectively "frozen" in its initial condition, as reaction or relaxation rates are negligible compared to the flow rate. Frozen flow is particularly relevant in high-speed or hypersonic flows, where non-equilibrium effects become significant.

Fig. 2.6 illustrates the flight paths of the Apollo Command Module and the Shuttle Orbiter during reentry. The reentry trajectory spans a broad range of velocities and altitudes, making accurate trajectory prediction extremely challenging. As expected, the atmospheric density changes dramatically with altitude, significantly influencing the aerodynamic forces acting on the vehicle. These forces, in turn, affect the vehicle's overall trajectory.

Atmospheric density is inversely related to temperature, and temperature variations directly impact the chemical composition of the surrounding air. As the vehicle descends, encountering higher velocities, the steep density and temperature gradients across shock waves become more pronounced. The faster the reentry vehicle, the sharper these gradients, which leads to complex shock-layer phenomena, including chemical reactions and ionization.

During reentry, the vehicle passes through various regions of the atmosphere where different chemical reactions occur, significantly altering the surrounding air's composition. For instance, at high velocities and temperatures, air molecules such as molecular oxygen and nitrogen dissociate, and ionization occurs, creating a plasma sheath around the vehicle. This plasma sheath influences the vehicle's electromagnetic properties, such as radio signal attenuation (blackout), which is critical for communication during reentry [16], [17].

These chemical effects must be considered carefully, as they impact not only the vehicle's aerodynamics but also thermal protection systems and onboard systems reliant on communication and navigation.

Fig. 2.6 demonstrates that as the reentry vehicle traverses different atmospheric chemistry regions, the resulting plasma and thermochemical effects become crucial factors in mission planning and vehicle design.

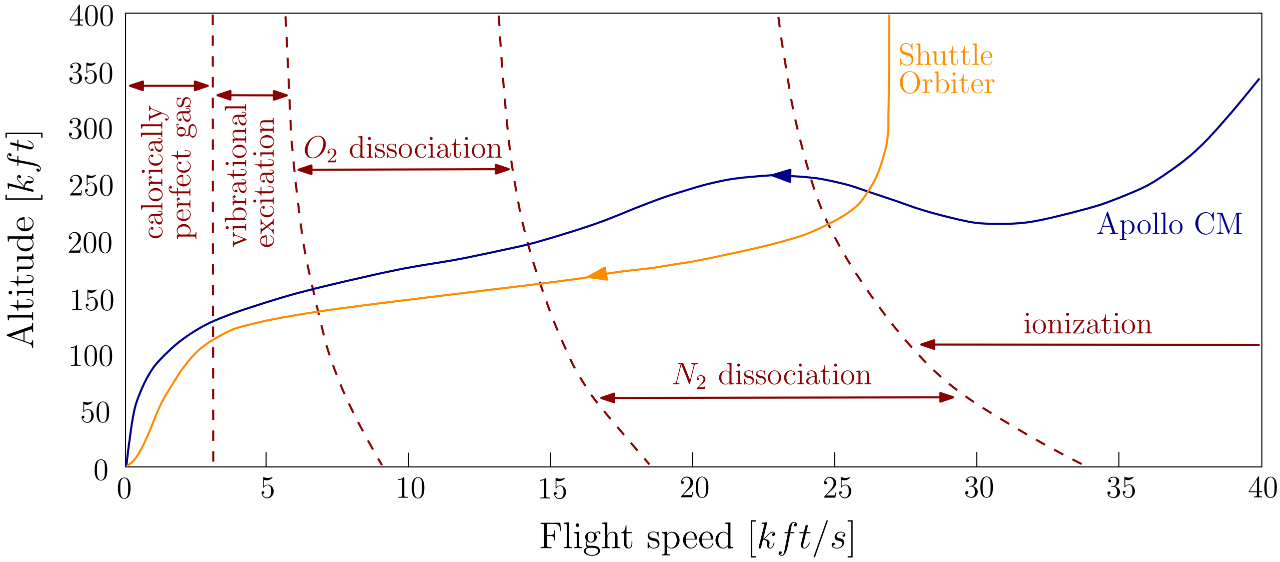


Figure 2.6: Velocity-altitude trajectories for different hypersonic flight systems

2.2 Optics Overview

The origins of optics trace back to the earliest foundations of science, as ancient Greek philosophers like Pythagoras, Democritus, Empedocles, Plato, Aristotle, and others formulated early theories on the nature of light [18]. These early thinkers debated whether vision was the result of ”rays” emitted from the eye or light entering it from external sources. In the 17^{th} century, Isaac Newton advanced the study of optics with his proposal that ”rays of light are very small bodies emitted from shining substances.” Around the same time, Christian Huygens proposed an alternative description, suggesting that light is a ”wave motion” spreading out from its source in all directions [19]. Huygens also correctly concluded that light slows down upon entering a denser medium, a key insight distinguishing his wave theory from Newton’s particle theory.

The field of optics took a major step forward in the 19^{th} century with James Clerk Maxwell’s development of electromagnetic theory, which provided a comprehensive description of light as a form of electromagnetic waves. Maxwell's equations unified electricity, magnetism, and optics, offering a new perspective on visible light as part of a broader spectrum of electromagnetic waves. In the 20^{th} century, advancements in Quantum-Mechanics (QM) introduced the concept of *quantized* electromagnetic energy, revealing that electromagnetic energy can only be exchanged in discrete packets known as *photons*.

Today, the theories of Newton and Huygens are understood to complement one another, with Maxwell's theory describing the propagation of light as an electromagnetic wave and quantum theory addressing the interaction of light and matter, including the processes of absorption and emission [19]. Together, these theories form the dual wave-particle understanding of light, a cornerstone of modern optics. Fig. 2.7 shows the energy spectrum, this work will stay in the low visible light, ignoring the electron transitions.

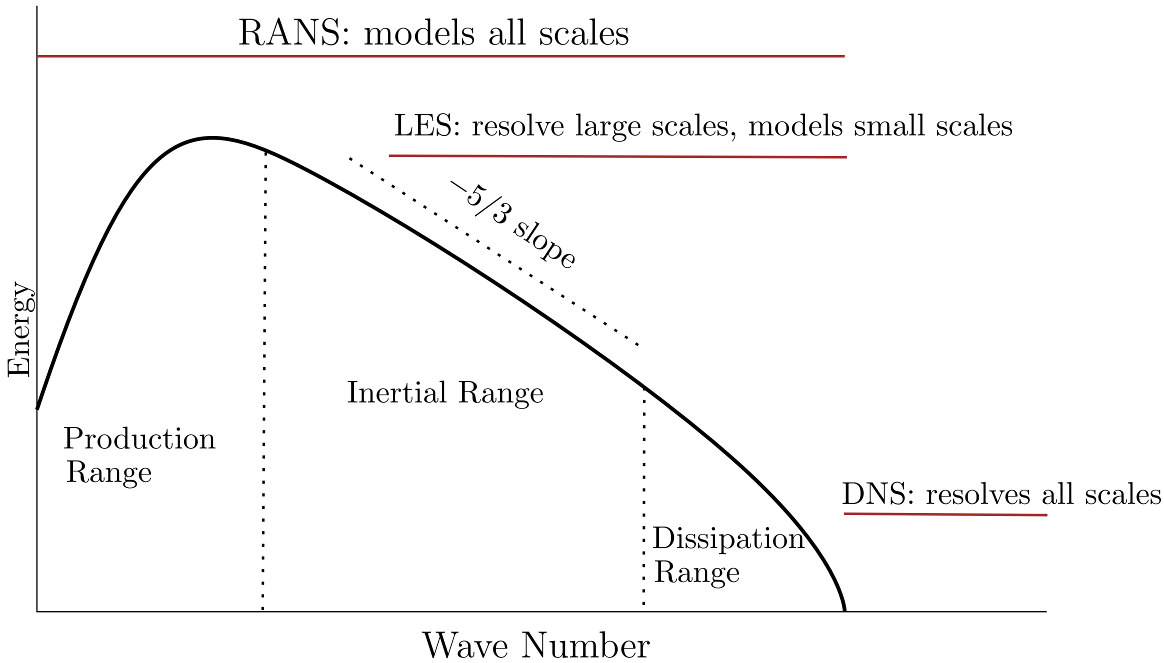


Figure 2.7: Energy spectrum

The frequencies of transitions between electronic levels lie in the visible and ultraviolet, the vibrational transition in the infrared, and the rotation transition in the far infrared [20].

As electromagnetic waves travel through a medium, these waves induce a polarization and magnetization on the medium's molecules; leading to the formation of dipoles. These dipoles generate their own electric and magnetic waves, representing bound charges and bound currents, respectively. Consequently, the newly formed wave arising from the combined effects of polarization and magnetization, interacts with the incident wave of identical frequency but varying propagation velocities [21], [22]. This varying in propagation velocity can be quantify by the index of refraction given in eq. (2.11). Essentially, the index of refraction is an essential property in optics that determines the speed by which light travels through a medium other than vacuum.

$$n = \frac{c_0}{c_m} \equiv \sqrt{\frac{\mu_m \epsilon_m}{\mu_0 \epsilon_0}} \approx \sqrt{\frac{\epsilon_m}{\epsilon_0}} \tag{2.11}$$

Fig. 2.8 shows a picture of a material bending a beam of light. Inside the material the speed of light is lower than outside the material. The same effect takes place when a beam of light travels across a denser medium such as a boundary layer.

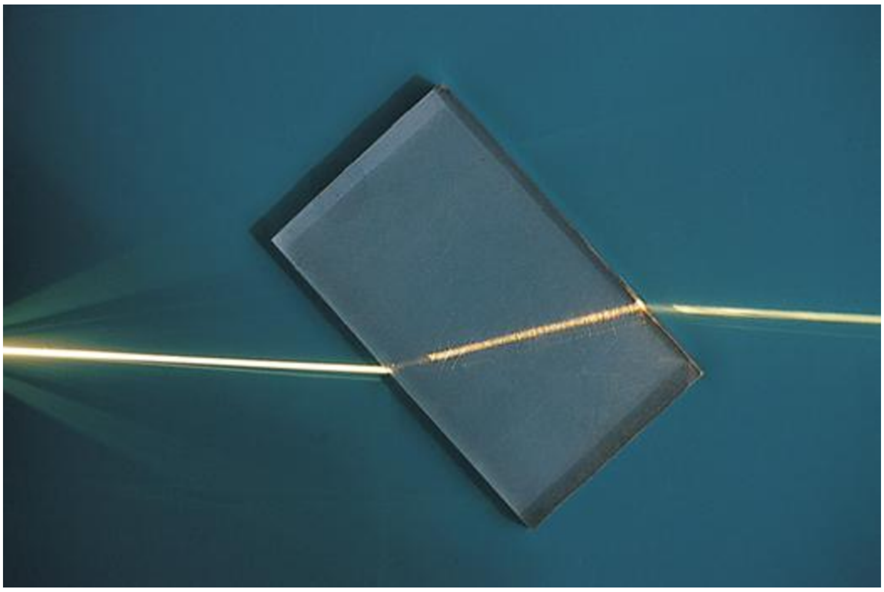


Figure 2.8: A ray of light refracted in a plastic block

The speed light in a thinner medium is larger than in a denser medium [23]. Thus the speed of light in a medium is inversely proportional to the medium's density eq. (2.12).

$$\rho_m \propto \frac{1}{c_m} \tag{2.12}$$

As a vehicle moves through a medium, it generates pressure waves. When the vehicle travels at speeds faster than the speed of sound in that medium, these pressure waves cannot move ahead of the vehicle and start to compress and bunch together near the leading edge. This compression creates a sudden change in pressure and density known as a shock wave. Shock waves are characterized by an abrupt and nearly discontinuous change in the properties of the medium, such as pressure, temperature, and density, which travel through the medium at supersonic speeds. fig. 2.9 illustrates how the pressure wave expansion changes with varying Mach numbers. Shock waves begin to form when the Mach number exceeds unity, resulting in different fluid properties, such as density, before and after the shock. This density gradient affects the index of refraction.

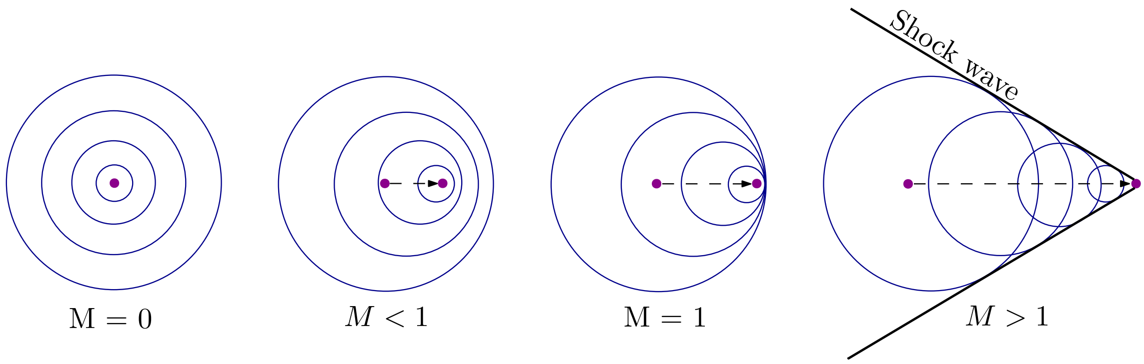


Figure 2.9: Shock waves formation

Optical wave propagation through an inhomogeneous medium where the index of refraction varies, results in wavefront deformation, as illustrated in fig. 2.10. This deformation can significantly degrade the performance of optical systems. However, it can also be leveraged as a visualization technique, such as the Schlieren method, used by experimental fluid researchers. This method exploits variations in the refractive index to visualize fluid flow, density gradients, and other phenomena in transparent media.

It is important to distinguish between the term "near field" as used in this context and its use in physical optics. In the current context, "near field" refers to aberrating fluid structures that are located very close to the exit aperture of the optical system. In contrast, in physical optics, the "near field" refers to a region where the Fresnel approximations are applicable, but the Fraunhofer approximations are not [24].

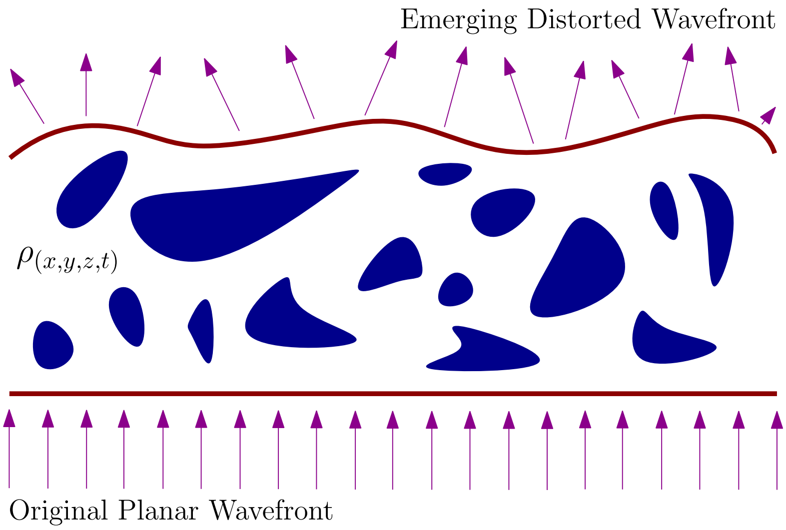


Figure 2.10: Planar optical wavefront distorted by a variant index of refraction flow-field

In this context, short-path (near-field) aberration is commonly referred to as *aero-optics*, which deals with the optical distortions caused by turbulent airflows near the optical system, such as those experienced by airborne or ground-based sensors. On the other hand, long-path aberrations, which occur over extended distances through the atmosphere, are generally referred to as "atmospheric propagation" effects. These long-path distortions are typically due to atmospheric turbulence, temperature gradients, and other environmental factors that cause fluctuations in the index of refraction over the optical path [24].

Optical aberrations such as the ones generated by turbulent flows, can be measured in terms of the OPL, given in eq. (2.13). The OPL is a measure of the effective distance that light travels in a medium, taking into account both the physical distance and the refractive index of the medium. The OPL determines the phase change that light undergoes as it propagates through a medium [25]. In other words, it is the "effective" distance light travels when considering how different materials slow down the speed of light relative to its speed in a vacuum.

$$OPL(t,x) = \int_{y_1}^{y_2} n(t,x,y)dy \tag{2.13}$$

Instead of measuring the absolute OPL, only the relative difference in OPL across the aperture is important. The OPD is given by eq. (2.14). The OPD directly reflects the variations of the wave-front phase errors [26].

$$OPD(t,x) = OPL(t,x) - \overline{OPL}(t) \tag{2.14}$$

Note: $\overline{(\ )}$ denotes the spatial average over the aperture.

Fluctuation in the fluid fields are not the only cause of distortion in Aero-Optical (AO). At high speed flows the nonequilibrium effects play an important role. Mackey [27], [28], [29] demonstrated that there is a discrepancy in the AO such as the index of refraction between a real and a perfect gas.

The discrepancy is due to polarizability, which refers to a molecule's tendency to develop a temporary dipole moment when exposed to an electric field. When an electric field is applied, the positively charged nucleus and negatively charged electron cloud experience opposing forces, causing a slight shift in their positions relative to each other. This shift creates an induced dipole moment.

In other words, polarizability measures how easily a molecule's electron cloud can be distorted or shifted in response to an electric field [30]. Larger, more spread-out electron clouds are generally more polarizable because they are less tightly bound to the nucleus, making them easier to distort. Fig. 2.11 illustrates the polarizability of a molecule when an electric field is applied to it.

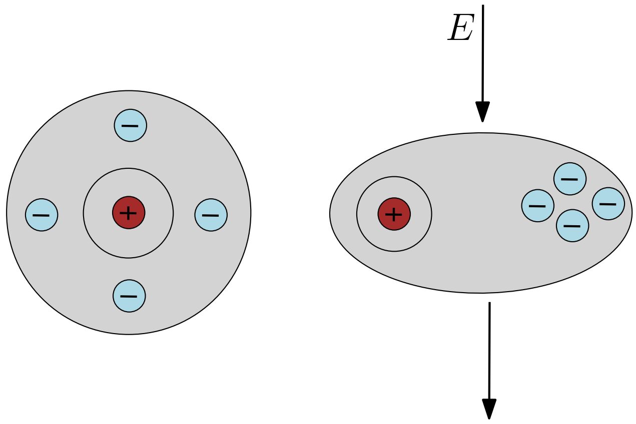


Figure 2.11: Polarizability, the tendency of a molecule to form a dipole moment when an electric field is applied

Polarizability influences the index of refraction by affecting how the material interacts with Electromagnetic (E&M) waves (light). When light enters a material, the oscillating electric field of the light wave induces a temporary dipoles in the atoms, or molecules in the material. This induced dipoles create their own electric field, which interfere with the incoming light wave, causing it to slow down as it propagates through the material. The degree of slowing down is influenced by the material's polarizability.

The polarizability may be usually separated into three distinct parts: electronic, ionic and dipolar as shown in fig. 2.12. The electronic polarizability arises from the displacement of the electron shell relative to the nucleus. The ionic polarizability arises from the displacement of charged ions with respect to other ions. The dipolar polarizability arises from molecules with a permanent electric dipole moment that can change orientation in an applied field [30].

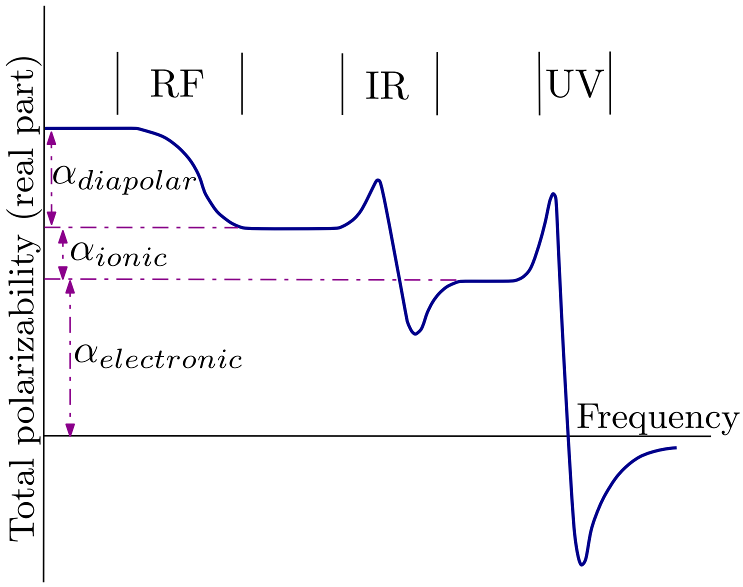


Figure 2.12: Frequency of the several contributions to polarizability

It has to be noted that polarizability it is a tensor but for cases of weak electric fields the rotational invariant polrizability eq. (2.15), which is one-third of the second-rank polarizability tensor is a good approximation. For high electric fields (non-linear optics), as well as electric multipole moments, the higher-order polarizabilities (hyper-polarizabilities), become important [31].

$$\alpha = \frac{\alpha_{xx} + \alpha_{yy} + \alpha_{zz}}{3} \tag{2.15}$$

Some papers provide the polarizability in Centimeter-Gram-Second units (CGS) and call it *volumetric* polarizability, some other used International System of Units (SI) and call it *atomic* polarizability. eq. (2.16) can be used to convert units between CGS and SI.

$$\alpha^{SI}_{[Fm^2]} = \frac{4\pi\varepsilon_0}{10^6}\alpha^{CGS}_{[cm^3]} \tag{2.16}$$

Traditionally polarizability is thought to be a function of the frequency of the E&M wave traveling across the medium. However, this is not the case; Hohm [31], [32] showed a temperature dependency on polarizability using interferometric measurements.

Rychlewski [33] showed a dependency in the rotational and vibrational quantum numbers in the polarizability constant. Kerl [34] used these previous results to develop a formulation for a temperature dependence polarizability. Kerl developed a curve fitting for the polarizability at different frequencies and temperatures eq. (4.1).

Tropina [35], [36] and Anaya [37] have demonstrated an influence of vibrational energy modes on polarizability. Kerl [31] and Buldakov [38], [39], [40] have demonstrated a temperature dependency in the polarizability. As expected in hypersonic flows, these phenomena are very important.

The polarizability is then use to calculate the index of refraction, eq. (2.17a) is valid for a dilute medium. For a dense medium the *Clausius-Mossotti* eq. (2.17b) can be used [23], [41], [42].

$$n - 1 = \frac{1}{2\epsilon_0} \sum_{i=1}^{N} \alpha_i N_i \tag{2.17a}$$

$$\frac{n^2 - 1}{n^2 + 2} = \frac{1}{3\epsilon_0} \sum_{i=1}^{N} \alpha_i N_i \tag{2.17b}$$

Some authors [43], [44], [45], [46], [47] use the G-D eq. (2.18b) constant to calculate the index of refraction eq. (2.18c). This constant was first formulated by *Gladstone* & *Dale* in 1864 and it is heavily used in the field of mineralogist [43].

$$K_s = \frac{\alpha_s^{SI}}{2\epsilon_0} \left(\frac{N_A}{M_s} \right) \left[\frac{m^3}{kg} \right] \tag{2.18a}$$

$$K_{GD} = \frac{1}{\rho} \sum_{s=1}^{N} K_s \rho_s \tag{2.18b}$$

$$n = \rho K_{GD} + 1 \tag{2.18c}$$

Some studies on optical propagation with applications to hypersonic reentry were performed by Gupta [47]. In this work, Gupta used the G-D constant to compare two methods (Snell's law and the optical Lagrange equation) for predicting the propagation of an optical signal from the Apollo 11 capsule during reentry. Gupta demonstrated that both techniques (Snell's law and optical Lagrange equations) matched well. It was observed in this study that the contribution to the refractive index by diatomic nitrogen and oxygen is greater upstream than downstream.

2.3 Quantum Mechanics Overview

Unlike Newtonian mechanics, Maxwell's electrodynamics, or Einstein's relativity, QM was not developed by a single individual [48] but by multiple physicists, including Planck, Bohr, Heisenberg, Schrödinger, Dirac, and others. To better understand the connection between hypersonic flows and optics, an overview of QM is necessary.

As previously explained in section 2.1, in a hypersonic flow, the *Hamiltonian* eq. (2.7) of a system must be modified. In molecular spectroscopy, the Born-Oppenheimer (BO) approximation is used to represent the total energy of the system [49]. The BO assumes that the wave function of an atomic nuclei and electrons in a molecule can be treated separately, base on the fact that the nuclei are much heavier than the electrons [50], [51] and it is shown in eq. (2.19).

$$\psi_{total} = \psi_{elec}\,\psi_{vib}\,\psi_{rot} \tag{2.19a}$$

$$\mathcal{H} = \varepsilon_{elec} + \varepsilon_{vib} + \varepsilon_{rot} \tag{2.19b}$$

Note ψ is the wave function and in order to calculate it, the Schrödinger's equation eq. (2.20) needs to be solved.

$$\hat{H}\,|\psi\rangle = i\hbar\frac{\partial}{\partial t}\,|\psi\rangle \tag{2.20}$$

The vibration-rotation energy for a diatomic molecules approximated up to the 2nd order is shown in eq. (2.21), and the general form using using the *Dunham* energy coefficients is shown in eq. (2.22) [52].

$$E_{v,J} = \underbrace{\omega_e\left(v+\frac{1}{2}\right) + B_e J\left(J+1\right)}_{\text{Harmonic}} - \overbrace{\omega_e x_e\left(v+\frac{1}{2}\right)^2 - D_e J^2\left(J+1\right)^2}^{\text{Anharmonic}} - \underbrace{\alpha_e\left(v+\frac{1}{2}\right) J\left(J+1\right)}_{\text{Vibration-rotation interaction}} \tag{2.21}$$

$$E_{vJ} = \sum_{m,n} Y_{mn}\left(v+\frac{1}{2}\right)^m J^n\left(J+1\right)^n \tag{2.22}$$

Some textbooks [53] use only the *Harmonic* terms from eq. (2.21) to define the *Hamiltonian*, which is a good approximation near the ground state. The vibrational energy for the *harmonic* energy equation is given in eq. (2.23a) for the vibrational term and in eq. (2.23b) for the rotational term. The additional terms come from perturbation theory and are called the *anharmonic* terms [48].

$$\varepsilon_{vib} = \omega_e\left(v+\frac{1}{2}\right)\left[\frac{1}{cm}\right] \tag{2.23a}$$

$$\varepsilon_{rot} = B_e J\left(J+1\right)\left[\frac{1}{cm}\right] \tag{2.23b}$$

Tab. 2.1 and tab. 2.2 provide the spectroscopy constants used to calculate eq. (2.21). These constants are given in wavenumber units, which are sometimes called *Kayser* units and are often used in spectroscopy [54]. These units represent the number of wavelengths per centimeter [55].

Molecule	$\omega_e\ \left[cm^{-1}\right]$	$\omega_e x_e\ \left[cm^{-1}\right]$	$\omega_e y_e\ \left[cm^{-1}\right]$	$B_e\ \left[cm^{-1}\right]$	$\alpha_e\ \left[cm^{-1}\right]$	$D_e\ \left[cm^{-1}\right]$	$ZPE\ \left[cm^{-1}\right]$
$NO+$	2376.72	16.255	−0.01562	1.997195	0.018790	5.64E−6	1184.332
$N2+$	2207.0115	16.0616	−0.04289	1.93176	0.0181	6.10E−6	1099.213
$O2+$	1905.892	16.489	0.02057	1.689824	0.019363	5.32E−6	948.906
NO	1904.1346	14.08836	0.01005	1.704885	0.0175416	0.54E−6	948.647
$N2$	2358.57	14.324	−0.00226	1.99824	0.017318	5.76E−6	1175.778
$O2$	1580.161	11.95127	0.0458489	1.44562	0.0159305	4.839E−6	787.380
$H2$	4401.21	121.33	−	60.853	3.062	0.0471	2179.2051

Table 2.1: Spectroscopy energy constants for ground level [56], [57]

The reduced mass (μ) was calculated using eq. (2.24a) and the molecular force (k) constants are calculated using eq. (2.24b). The molecular force constant measures the stiffness of a chemical bond.

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \tag{2.24a}$$

$$k = \mu\omega^2_{\left[\frac{rads}{sec}\right]} = \mu\left(100 \times 2\pi c_0 \nu_{[cm^{-1}]}\right)^2 \left[\frac{N}{m}\right] \tag{2.24b}$$

Molecule	r_e $[\mathring{A}]$	$\mu \times 10^{-26}$ $[kg]$	k $[N/m]$
$NO+$	1.06322	1.240	2485.629
$N2+$	1.11642	1.628	2009.857
$O2+$	1.1164	1.328	1712.066
NO	1.15077	1.240	1595.422
$N2$	1.09768	1.162	2295.375
$O2$	1.20752	1.328	1176.865
$H2$	0.74144	0.0836	575.174

Table 2.2: Spectroscopy constants for ground level [57]

Kayser units can be easily converted to energy units of *Joules* using eq. (2.25a) and to *electronvolts* using eq. (2.25b).

$$[J] = \left[\frac{1}{cm}\right] \times c_0 \left[\frac{m}{s}\right] \times 2\pi\hbar\ [J\ s] \times 100 \tag{2.25a}$$

$$[eV] = \left[\frac{1}{cm}\right] \times c_0 \left[\frac{m}{s}\right] \times 2\pi\hbar\ [J\ s] \times 100 \times \frac{1}{q_e}\ [eV] \tag{2.25b}$$

The Zero Point Energy (ZPE) was calculated using eq. (2.26) provided in [56]. The ZPE is the lowest possible energy that a quantum mechanical system may have. It has to be noted that this is only possible to be obtained for diatomic molecules, because monatomic molecules only have translational energy modes [58].

$$ZPE = \frac{\omega_e}{2} - \frac{\omega_e x_e}{2} + \frac{\omega_e y_e}{8} + \frac{B_e}{4} + \frac{\alpha_e \omega_e}{12 B_e} + \frac{\alpha_e^2 \omega_e^2}{144 B_e^3} \tag{2.26}$$

Some authors refer to the constants in tab. 2.1 as the *Dunham* coefficients [59] ($Y_{ij} = Y_{vJ}$) [60]. However, in the literature, these coefficients may also be referred to as the *Dunham* potential coefficients, which are not the same. Ogilvie [61] called the Y_{vJ} energy-coefficients and a_m the *Dunham* potential-coefficients. These conventions will be used for the remainder of this work.Eq. (2.27) provides a relationship between the energy coefficients and the spectroscopy constants found in tab. 2.1.

$$\begin{array}{llll} Y_{00} = ZPE & Y_{02} = D_e & Y_{11} = -\alpha_e & Y_{30} = \omega_e y_e \\ Y_{01} = B_e & Y_{10} = \omega_e & Y_{20} = -\omega_e x_e & \end{array} \tag{2.27}$$

The *Dunham* potential coefficients are found using eq. (2.28a) [61], equations (2.28b) and (2.28c) [62], and eq. (2.28d) [63].

$$a_0 = \frac{\omega_e^2}{4B_e} \tag{2.28a}$$

$$a_1 = -\left(1 + \frac{\alpha_e \omega_e}{6B_e^2}\right) \tag{2.28b}$$

$$a_2 = \frac{5}{4}a_1^2 - \frac{2}{3}\left(\frac{\omega_e x_e}{B_e}\right) \tag{2.28c}$$

$$a_m = \left(\frac{12}{a_1}\right)^{m-2} \left(2^{m+1} - 1\right) \left(\frac{a_2}{7}\right)^{m-1} \prod_{k=0}^{m-3} \frac{1}{m+2-k} \tag{2.28d}$$

Tab. 2.3 shows the *Dunham* potential coefficients for multiple diatomic constants using eq. (2.28).

Molecule	$a_0 \times 10^5$	a_1	a_2	a_3
$NO+$	7.071	-2.866	4.842	-6.009
$N2+$	6.304	-2.784	4.146	-4.536
$O2+$	5.374	-3.154	5.929	-8.189
NO	5.317	-2.915	5.114	-6.592
$N2$	6.960	-2.705	4.367	-5.179
$O2$	4.318	-3.008	5.795	-8.205
$H2$	0.795	-1.606	1.897	-1.645

Table 2.3: Potential *Dunham* coefficients, eq. (2.28)

Buldakov [38] postulated a formulation for the polarizability of diatomic molecules, assuming that near the equilibrium internuclear distance, the polarizability can be expressed as a power series eq. (2.29).

$$\alpha(r, \omega) = \alpha_e + \alpha_e' x + \frac{1}{2}\alpha_e'' x^2 + \frac{1}{6}\alpha_e''' x^3 \tag{2.29}$$

Buldakov then expressed the polarizability of diatomic molecules using eq. (2.30). The full expansion of this equation can be found in chapter A. Tab. 2.4 shows the polarizability derivatives used in this method, and it requires the *Dunham* potential-coefficients.

$$\alpha(\omega, T) = \sum_{vJ} p_{v,J}(T) \langle v, J | \alpha^{(i)}(r, \omega) | v, J \rangle \tag{2.30}$$

Molecule	$\alpha_e \times 10^{-30}$ $[m^3]$	$\alpha'_e \times 10^{-30}$ $[m^3]$	$\alpha''_e \times 10^{-30}$ $[m^3]$	$\alpha'''_e \times 10^{-30}$ $[m^3]$
$H2$	0.7849	0.90	0.49	-4.6
$N2$	1.7801	1.860	1.20	-4.60
$O2$	1.6180	1.760	3.40	23.70

Table 2.4: Polarizability derivatives [38]

Eq. (2.30) requires calculating the Boltzmann distribution (eq. (2.31)). However, the Boltzmann distribution assumes that the system is in thermal equilibrium.

A system is in thermodynamic equilibrium when the entropy of the system has reached its maximum possible value. Entropy is a measure of the number of possible microscopic configurations that correspond to the macroscopic state of the system [64]. At low vibrational and rotational quantum numbers, the entropy is lower due to fewer accessible states in which energy can be distributed. In contrast, at higher quantum numbers, there are more possible "energy buckets" available for energy distribution, leading to increased entropy.

The energy distribution of a system in thermal equilibrium can be mathematically described by the Boltzmann distribution. This distribution provides the probability of a system occupying a specific energy state, showing that states with higher energy are less probable but increasingly accessible as temperature rises [65]. Consequently, the Boltzmann distribution is foundational in explaining how temperature and entropy are related in a system at equilibrium, offering insight into molecular behavior at varying energy levels.

The Boltzmann distribution, which describes the particle distribution over different energy states at thermal equilibrium, is applied to calculate the energy levels as a function of the vibrational quantum number. This probability distribution is given in eq. (2.31).

$$p_{i,j} = \frac{g_j e^{-\beta E_{v,J}}}{Z} = \frac{(2J+1)\, e^{-\beta(\varepsilon_i+\varepsilon_j)}}{\sum_{v,J} (2J+1)\, e^{-\beta(\varepsilon_v+\varepsilon_J)}} \tag{2.31}$$

It should be noted that, in a system of harmonic oscillators, vibrational energy relaxes through a series of Boltzmann distributions. However, this behavior does not hold for anharmonic oscillators [66]. The assumption of harmonic behavior can be applied to the rigid rotor model [67]. Therefore, for the purposes of this approach, it is assumed that the system consists solely of harmonic oscillators. As a result, the anharmonic terms in the vibrational-rotational energy described by eq. (2.21) will be neglected.

2.4 Turbulence Overview

Feynman once stated, *"Turbulence is the most important unsolved problem of classical physics."* This statement remains true to this day. The problem of turbulence has attracted countless mathematicians, physicists, and engineers, yet it remains unsolved. Davidson et al. [68] provide an excellent historical overview of the field, tracing its development and the scientist who have contributed to the field of turbulence.

Defining turbulent flow is considerably more complex than defining compressible or hypersonic flow. Unlike the latter, which are based on measurable parameters such as Mach number or density variations, turbulent flow encompasses a broad spectrum of characteristics that resist simple classification. Tennekes [69] characterizes turbulent flow by identifying the following defining features:

- **Irregularity:** Turbulent flows are inherently chaotic and unpredictable in their detailed behavior.

- **Diffusivity:** Enhanced mixing and transport of momentum, heat, and species occur due to turbulence.
- **Large Reynolds numbers:** Turbulence typically arises in flows with high Reynolds numbers, where inertial forces dominate over viscous forces.
- **Three-dimensional vorticity fluctuations:** Turbulence involves complex, unsteady, and three-dimensional vortex structures.
- **Dissipation:** Energy injected at larger scales cascades to smaller scales and is ultimately dissipated as heat due to viscosity.
- **Continuum:** Despite chaotic fluctuations, turbulent flows are governed by the principles of continuum mechanics.

Pope [70] and Wilcox [71] similarly refrained from providing a strict definition of turbulence. Instead, they opted to describe its characteristics, aligning with Tennekes' approach [69]. This reflects the inherent complexity and multifaceted nature of turbulence, which defies a singular definition.

Davidson [72], [73] offers a conceptual perspective, defining an eddy as a "blob of vorticity" and describing turbulence as a "sea of eddies." In this view, vortices within the turbulent flow are stretched and deformed by the velocity field, which, in turn, is governed by the instantaneous distribution of vorticity. This interplay highlights the dynamic and self-sustaining nature of turbulence, where vorticity and velocity fields are intricately linked.

Indeed, turbulence remains one of the most challenging and elusive topics in fluid mechanics. Despite decades of research, its complexity continues to captivate and challenge mathematicians, physicists, and engineers alike.

Fig. 2.13 captures the atmospheric turbulence of Jupiter as observed by National Aeronautics and Space Administration (NASA)'s Juno spacecraft. This stunning image reveals the dynamic and chaotic patterns within Jupiter's upper atmosphere, showcasing a complex interplay of swirling clouds, vortices, and turbulent flows. The visible turbulence is driven by the planet's rapid rotation, strong convection currents, and immense atmospheric pressure, which collectively give rise to the characteristic banded appearance and intricate flow structures.

These turbulent flows exemplify the multi-scale nature of turbulence, where energy cascades from large, organized structures—such as Jupiter's iconic bands and cyclones—to smaller, chaotic eddies and vortices. The intricate patterns demonstrate the non-linear behavior of fluids under turbulent regimes, providing a planetary-scale example of phenomena also observed in laboratory and terrestrial fluid systems.

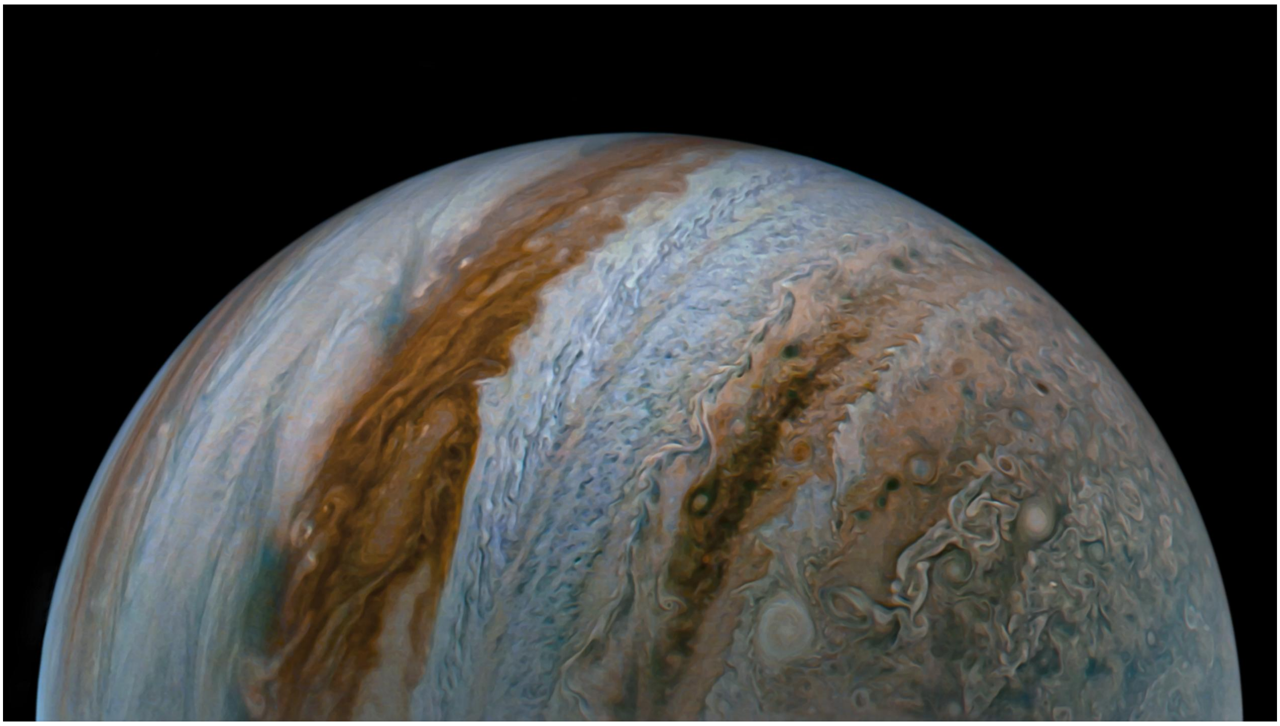


Figure 2.13: Jupiter's view captured by Juno spacecraft

Fig. 2.14 illustrates turbulence generated in coffee during the mixing process, showcasing the chaotic and intricate flow structures that arise. The turbulence in coffee is not only visually striking but also plays a significant role in enhancing the extraction of soluble compounds and influencing the flavor profile. The mixing ensures a uniform distribution of dissolved substances such as oils, acids, and sugars, which contribute to the overall taste and aroma of the coffee [74], [75].

This image captures the transition from laminar to turbulent flow as the milk is introduced, with large-scale swirling motions breaking down into smaller, more chaotic structures. These turbulent interactions promote rapid mixing and facilitate efficient mass transfer, ensuring that the milk and coffee achieve a homogeneous blend. Such flow dynamics demonstrate the multi-scale nature of turbulence, where energy is transferred from larger eddies to smaller ones in a cascading process.

Figure 2.14: Turbulence in coffee

One of the most important characteristics of turbulence is the presence of different scales within the flow. In 1922 Richardson introduced the energy cascade concept "Big whorls have little whorls that feed on their velocity; and little whorls have lesser and so on to viscosity." Fig. 2.15 shows the transfer of energy from the largest scales to the smaller scales, this process is achieved by the "vortex-stretching" [73].

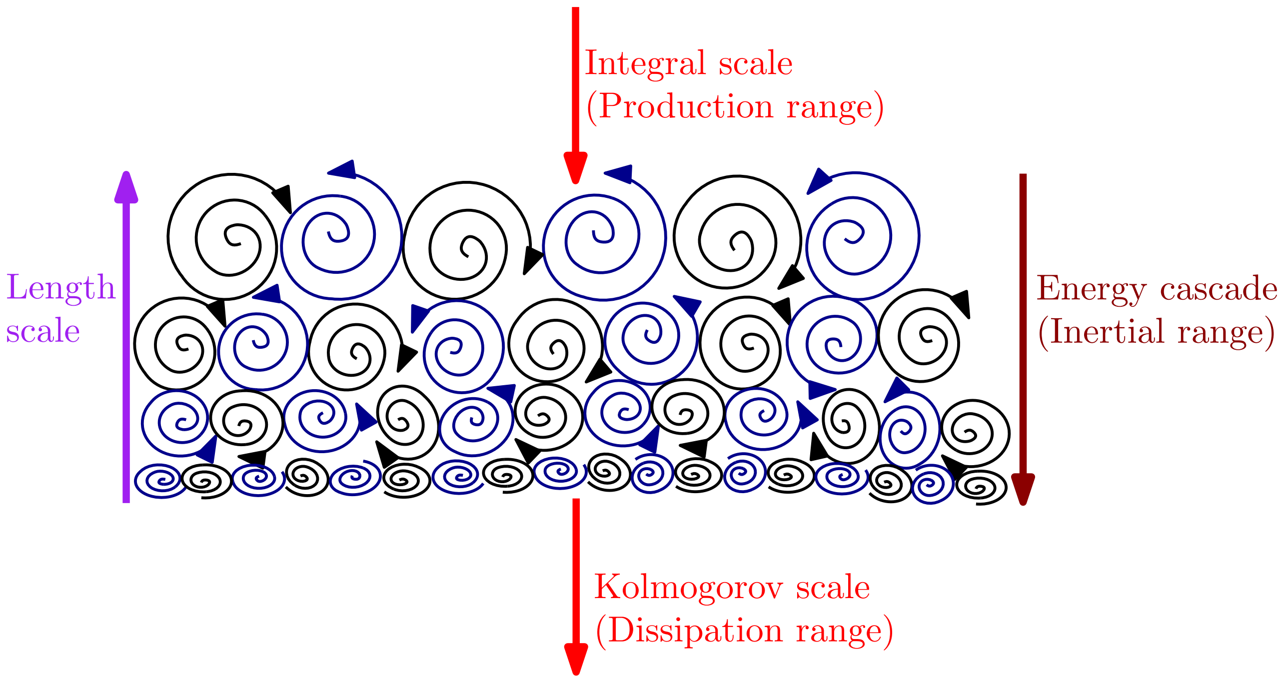


Figure 2.15: Energy cascade

These scales represent the hierarchical organization of turbulent eddies, which vary in size and energy transfer mechanisms. The largest eddies extract kinetic energy from the mean flow in the region known as the production region. The characteristic length scales in this region, known as the integral length scales, are comparable to the flow dimensions. This process is highly anisotropic, meaning the flow properties depend on direction, and it is primarily influenced by transport and mixing rather than viscosity.

In the inertial range, the intermediate eddies are characterized by the Taylor length scale. Turbulence in this region is isotropic, indicating uniformity in all directions, but these eddies are dynamically independent of both the largest eddies in the production region and the smallest eddies in the dissipation range. Eddies in the inertial range transfer energy from larger scales to smaller scales through a cascade process. Their behavior is commonly described using the turbulent dissipation rate (ε) and the wave number (κ), which inversely relates to the eddy size. In this range, the influence of viscosity is negligible, and the turbulence is governed by inertial forces.

The smallest eddies exist in the dissipation range, where they exhibit universal characteristics independent of the flow geometry or boundary conditions. These eddies receive energy transferred from larger eddies and dissipate it as heat through the fluid's molecular viscosity. The length scales of these eddies are described by the Kolmogorov scales, which depend on the kinematic viscosity (ν) and the turbulent dissipation rate (ε). These eddies are isotropic due to the dominance of viscosity at these scales, which smoothens out directional variations. The smallest-scale motions are entirely governed by dissipation, making them crucial for understanding energy conversion in turbulent flows.

Fig. 2.16 illustrates the energy spectrum of turbulence, which organizes the energy distribution across different scales within a turbulent flow. This spectrum is crucial for understanding the underlying mechanisms and characteristics of turbulence.

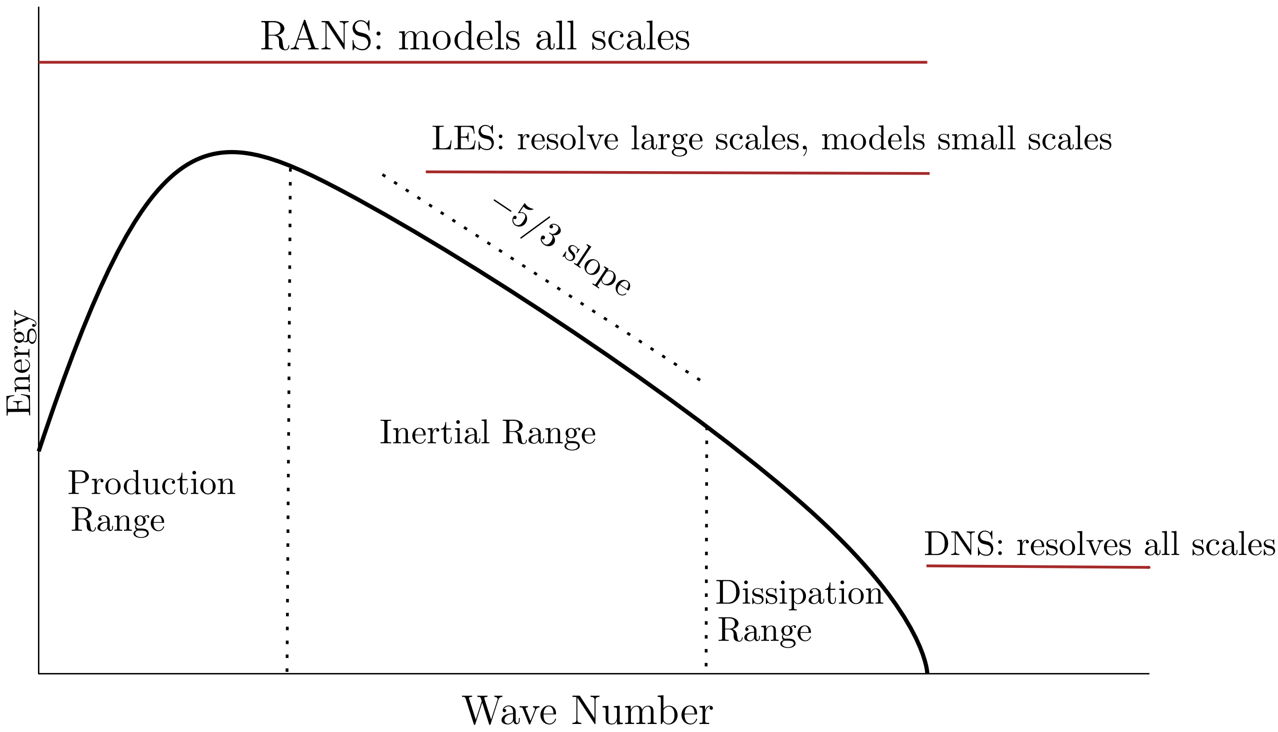


Figure 2.16: Energy spectrum

The energy spectrum is divided into three primary regions: the production range, the inertial range, and the dissipation range. It is important to note that the presence of vortices is a necessary but not sufficient condition for turbulence. At high

Reynolds numbers, an energy spectrum with a characteristic slope of $-\frac{5}{3}$ emerges in the inertial range [76], providing a more accurate definition of turbulence.

Most turbulent flows found in nature are bounded [70]. Examples of internal flows include flow through a pipe, while examples of external flows include flow around an aircraft. The study of these flows is highly challenging because the presence of a solid wall imposes additional constraints that are not present in free-shear flows (such as jets and wakes) [69].

A common tool to understand and characterize a turbulent Boundary Layer (BL) is to use the Van-Driest transformation, as shown in eq. (2.32).

$$u^+ = \frac{\overline{u}}{u_\tau} \tag{2.32a}$$

$$y^+ = \left(\frac{\rho_w u_\tau}{\mu_w}\right) \tag{2.32b}$$

$$u_\tau = \sqrt{\frac{\tau_w}{\rho_w}} \tag{2.32c}$$

A turbulent BL can be divided into regions based on the fluid properties. Depending on the value of the y^+, different regions within the BL are defined. Tab. 2.5 shows the characteristics of these regions.

Regions	Location	Description
Viscous sublayer	$y^+ < 5$ $u^+ = y^+$	Layer dominated by viscous stress and not the Reynolds stress.
Buffer layer	$5 < y^+ < 30$	Region between the viscous sublayer and log-law region.
Log-law layer	$30 < y^+ < 50$ $u^+ = \frac{1}{\kappa}\ln\left(y^+\right) + B$	The log-law holds.
Outer layer	$y^+ > 50$	Direct effects of viscosity on the velocity are negligible.

Table 2.5: Wall regions and layers, table from [70]

Characterizing these regions in a flow is essential to understand the effects of the wall on the flow. As the flow moves away from the wall, the impact of the wall shear stress decrease, and the Reynolds stress takes over. The majority of turbulence generation occurs in the buffer layer; this is the region in which ejection and sweeps also take place [77] and where the bursting process is most active. Hairpins in the BL can be oriented at 45° and are primary candidates for transferring energy from small scales to large scales (inverse energy cascade) [73]. Robinson [77] explains the bursting process, which is responsible for the inverse energy cascade, in detail.

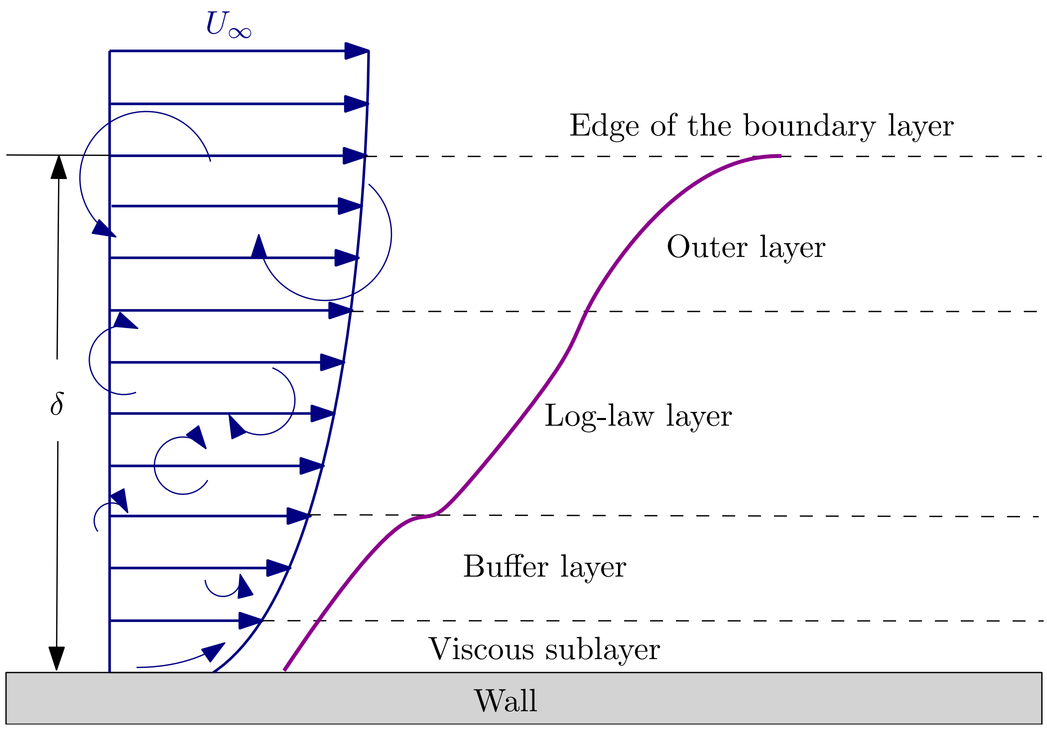


Figure 2.17: Turbulent boundary layer

When a beam of light travels across a BL, it interacts with the different regions within the boundary layer. The viscous sublayer is typically considered laminar; however, as the beam propagates across the boundary layer, turbulence becomes dominant. To accurately compute the AO properties, it is essential to account for all relevant scales in the turbulent flow. While optical effects are negligible at small scales in turbulence, they are significant at larger scales [78].

Both computational and experimental studies in BLs have demonstrated the presence of the energy cascade in wall-bounded flows. Marati et al. used results from a low-Reynolds-number Direct Numerical Simulation (DNS) to describe the characteristics of turbulence in different regions of the boundary layer [79].

Wyckham [80] conducted an experimental study on transonic and hypersonic BLs, showing that the largest scales in the boundary layer tend to dominate AO distortions. These large scales often begin to emerge in the log-law layer [81]. Additionally, most turbulence production occurs in the buffer layer [77].

Mackey et al. [27] performed an implicit Large Eddy Simulations (LES) for a $M = 4$ flow over an adiabatic flat plate without thermochemistry models and showed that AOs, such as the index of refraction and OPD properties, are affected by fluctuations in the flow field.

Miller [82] and others investigated the behavior of turbulence-induced aero-optical distortions in hypersonic boundary layers using DNS. Miller utilized the G-D to calculate the index of refraction and employed Gordeyev et al. [83] model to study the OPD at Mach numbers of 14. However; in this work, Miller did not consider nonequilibrium effects. This study fits into Quadrat II.

Passiatore et al. [84] conducted a study at $M = 12.48$ using DNS and demonstrated that most nonequilibrium effects occur within the buffer layer. Consequently, if an optical beam traverses the boundary layer, nonequilibrium effects will be predominant closer to the wall, peaking within the buffer layer, while becoming less significant in the log-law and outer layers.

In contrast, turbulent effects are negligible in the viscous sublayer but become increasingly important as the beam moves away from the wall. This is due to the larger eddies present in the log-law layer, where turbulence dominates. As a result, turbulent effects are more significant in this region compared to the viscous sublayer or buffer layer.

Chapter 3

Modeling

3.1 Computational Fluid Dynamics

The roots of CFD can be traced back to the 1950s and 1960s when researchers began using computers to solve Partial Differential Equations (PDEs) that describe fluid flow. This period saw the initial application of numerical analysis to solve simplified fluid problems.

Numerical analysis provides methods and algorithms that suitable for CFD such as Finite Volume Method (FVM), Finite Difference Method (FDM) and Finite Element Method (FEM) [85]. In the 1970s and 1980s, the field of CFD has been established as a distinct discipline, and the availability of more powerful computers allowed researches to address more complex and realistic problems. This development was driven by the aerospace community [86].

Between the 1980s to 1990s, CFD became a more widely and accessible with development of commercial CFD software such as ANSYS Fluent. During this period, the ability to simulate and analyze fluid flows using computational methods gained popularity in industries such as aerospace and automotive engineering. From the 1990s to the present, continuous advancements in numerical algorithms, grid generation techniques, and parallel computing have been made. Adaptive mesh refinement, unstructured grids, and High Performance Computing (HPC) have significantly improved the accuracy and efficiency of CFD simulations.

In recent years, there has been a push towards multidisciplinary simulations, integrating CFD with other disciplines such as structural mechanics, heat transfer and optics. This integration allows for solving more realistic and complex problems.

3.2 Hypersonic Modeling

Missing hypersonic modeling

A commonly used model to describe nonequilibrium effects is the Park $2T$ model [87], which assumes that the flow can be described by two temperatures. One temperature (T_{tr}) will describe the heavy-particle translational and molecular rotational energy and the other temperature (T_{vr}) will describe the molecular vibrational, electron translational, and electronic excitation energies.

Stanford University Unstructured (SU2)–NonEquilibriumMOdeling (NEMO) [88] solves the governing equations for viscous gases in thermochemical nonequilibrium to model continuum hypersonic flows. These equations are expressed compactly in conservation form in eq. (3.1) and expanded in eq. (3.2).

$$\mathcal{R}\left(\mathcal{U}, \nabla\mathcal{U}\right) = \frac{\partial \mathcal{U}}{\partial t} + \nabla \cdot \mathcal{F}^{c}\left(\mathcal{U}\right) - \nabla \cdot \mathcal{F}^{v}\left(\mathcal{U}, \nabla\mathcal{U}\right) - \mathcal{Q}\left(\mathcal{U}\right) = 0 \tag{3.1}$$

$$\mathcal{U} = \begin{Bmatrix} \rho_1 \\ \cdot \\ \cdot \\ \cdot \\ \rho_s \\ \rho\boldsymbol{u} \\ \rho e \\ \rho e^{ve} \end{Bmatrix}, \quad \mathcal{F}^c = \begin{Bmatrix} \rho_1 \boldsymbol{u}^T \\ \cdot \\ \cdot \\ \cdot \\ \rho_s \boldsymbol{u}^T \\ \rho\boldsymbol{u}\boldsymbol{u}^T + P\hat{I} \\ \rho h \boldsymbol{u}^T \\ \rho e^{ve}\boldsymbol{u}^T \end{Bmatrix}, \quad \mathcal{F}^v = \begin{Bmatrix} -\boldsymbol{J}_1 \\ \cdot \\ \cdot \\ \cdot \\ -\boldsymbol{J}_s \\ \boldsymbol{\sigma} \\ \boldsymbol{u}^T\boldsymbol{\sigma} - \sum_k \boldsymbol{q}^k - \sum_s \boldsymbol{J}_s h_s \\ -\boldsymbol{q}^{ve} - \sum_s \boldsymbol{J}_s e_s^{ve} \end{Bmatrix}, \quad \mathcal{Q} = \begin{Bmatrix} \dot{\omega}_1 \\ \cdot \\ \cdot \\ \cdot \\ \dot{\omega}_s \\ 0 \\ 0 \\ \dot{\Theta}^{tr:ve} + \sum_s \dot{\omega}_s e_s^{ve} \end{Bmatrix} \tag{3.2}$$

3.3 Turbulence Modeling

In 1963, Smagorinsky [89] introduced the LES model to simulate atmospheric air currents. The LES framework is grounded in the energy cascade model developed by Kolmogorov in the 1940s, which itself builds upon the earlier work of Richardson. The fundamental concept behind LES is to explicitly compute the large-scale motions, which are influenced flow geometry and are not universal, up to a prescribed length scale. The smaller scales, which tend to exhibit more universal characteristics, are modeled using a Sub-Grid Scales (SGS) model [70].

This separation between larger and smaller scales is typically achieved by spatially filtering the velocity field using a kernel [90]. The convolution kernel effectively splits the flow field into resolved large scales and modeled SGS components in an LES framework. Developing physically realistic SGS models requires a comprehensive understanding of both the physics and statistical properties of scale interactions between large and small scales in turbulent flows. SGS modeling is traditionally predicated on the hypothesis that if subgrid scales exist, the flow is locally turbulent in both space and time [91].

There are two primary SGS modeling strategies:

- **Functional modeling:** This approach focuses on the energetic impact of subgrid scales on resolved scales. It assumes that balancing energy transfer between scale ranges is sufficient to describe the subgrid-scale effects. Functional models often neglect backscattering, i.e., energy transfer from smaller scales back to larger scales [92].

- **Structural modeling:** This strategy attempts to reconstruct subgrid-scale information directly by generating a new field from the filtered field (e.g., velocity, density) that better approximates the unfiltered field. Structural models account for backscattering because they aim to reconstruct small-scale structures within the flow [92].

Functional models commonly represent the SGS stress tensor (Reynolds stress tensor) using eddy-viscosity closures. However, this approach linearizes the inherently nonlinear stress tensor, leading to the loss of some physical details. It is important to note that the famous closure problem originates from the convective terms in the Navier-Stokes (N-S) equations. Additionally, functional models are often criticized for being overly dissipative due to their nature of transferring energy from larger to smaller scales while failing to capture backscattering effects. In contrast, structural models incorporate backscattering and are typically nonlinear, allowing for a more accurate representation of small-scale flow structures.

Missing turbulence modeling

Chapter 4

Aero-Optics Tool

An open-source tool, HAOT, has been developed to perform this analysis. The tool is published on PyPI, the source code is available on GitHub, and the documentation can be found on Read the Docs. The current version of HAOT includes five modules:

- **Aerodynamics:** Implements commonly used aerodynamic functions.
- **Optics:** Implements optics equations used in hypersonic flow analysis.
- **Quantum Mechanics:** Implements commonly used quantum mechanical functions for hypersonic flow applications.
- **Constants:** Includes various constants, primarily spectroscopy constants, used across the different modules.
- **Conversion:** Provides several conversion functions used within the package.

Some Validations results are proved in sections below, the constants and conversions were validated by visual inspection.

4.1 Aerodynamics

Fig. 4.1 calculates the dynamic viscosity HAOT packages and compares the results against the Ambiance package, which is an implementation of the *U.S. Stan-*

dard Atmosphere, 1976 report.

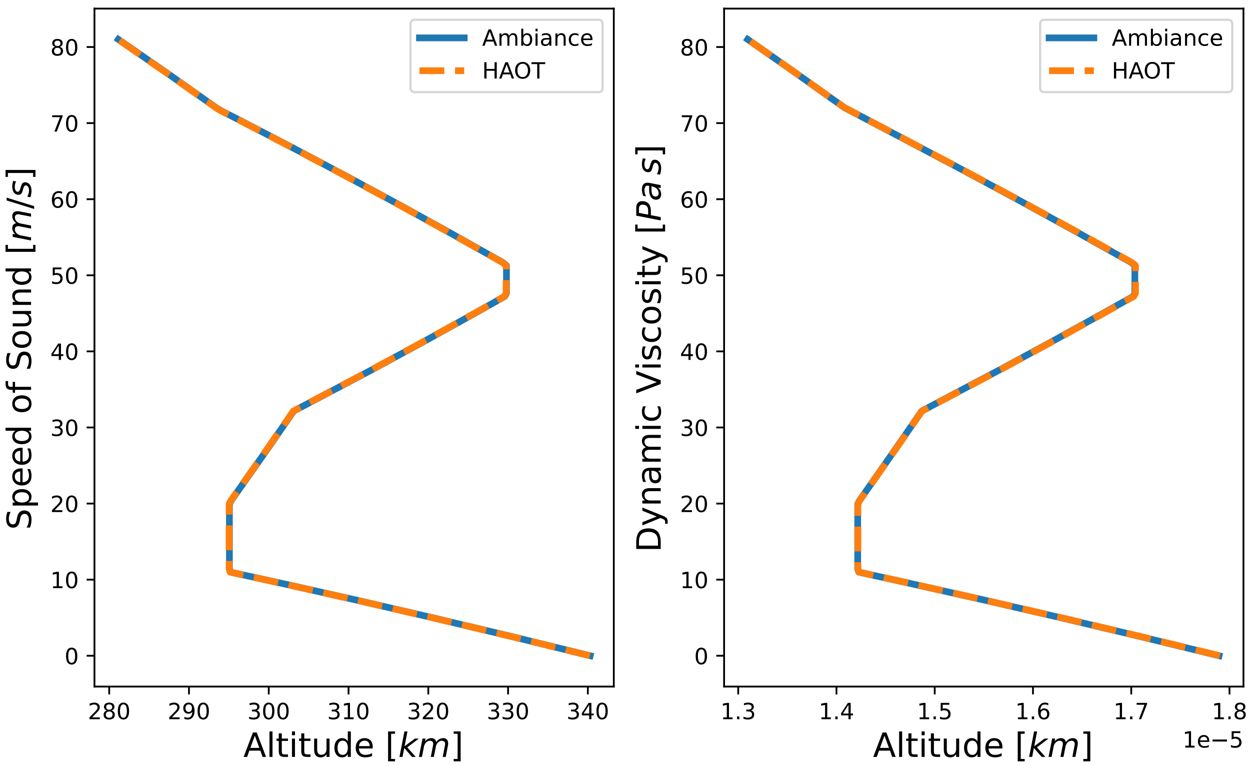


Figure 4.1: Ambiance Vs. HAOT

4.2 Optics

Fig. 4.2 shows the results of curve fitting developed by Kerl [34]. This fitting uses constants defined in tab. 4.1. These constants are accessible in the *constants* module from the HAOT package and results match published results by Kerl [34].

$$\alpha(\omega, T) = \frac{\alpha(0,0)}{1 - \left(\frac{\omega}{\omega_0}\right)^2} \left(1 + bT + cT^2\right) \tag{4.1}$$

Molecule	$\alpha_0 \times 10^{-30}\ \left[m^3\right]$	$\omega_0 \times 10^{16}\ \left[\frac{rads}{sec}\right]$	$b \times 10^{-6}\ \left[\frac{1}{K}\right]$	$c \times 10^{-9}\ \left[\frac{1}{K}\right]$
$N2$	1.7406	2.6049	1.8	–
$O2$	1.5658	2.1801	−2.369	8.687
Air	1.6970	2.47044	10.6	7.909

Table 4.1: Kerl fitting constants [34] eq. (4.1)

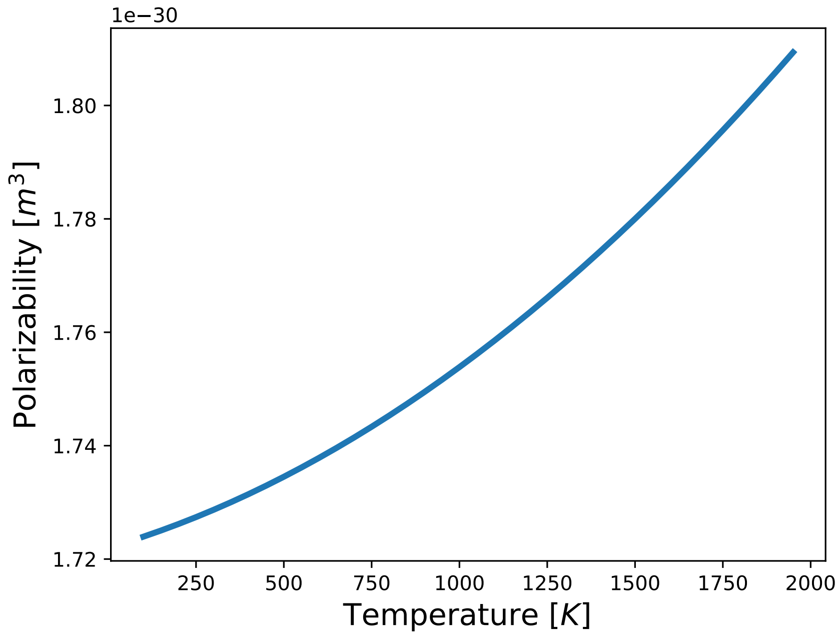


Figure 4.2: Polarizability of air for a wavelength of 633 $[nm]$

Figures 4.3 to 4.5 show the comparison between the HAOT package and the results published by Kerl [34]. These results show that the HAOT implementation matches the published results.

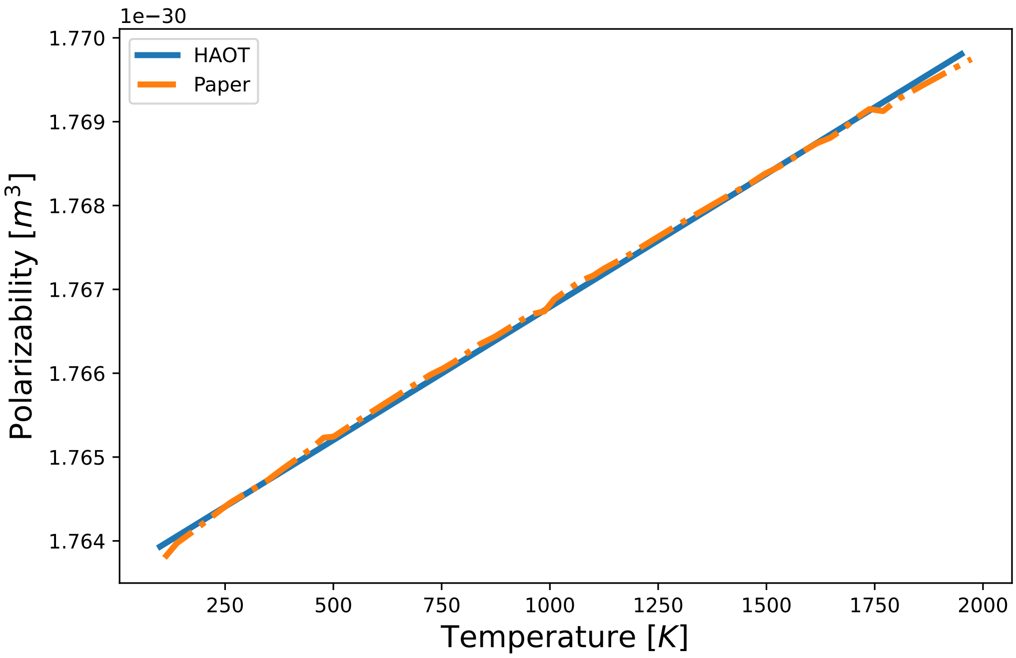


Figure 4.3: Polarizability of $N2$ for a wavelength of 633 $[nm]$

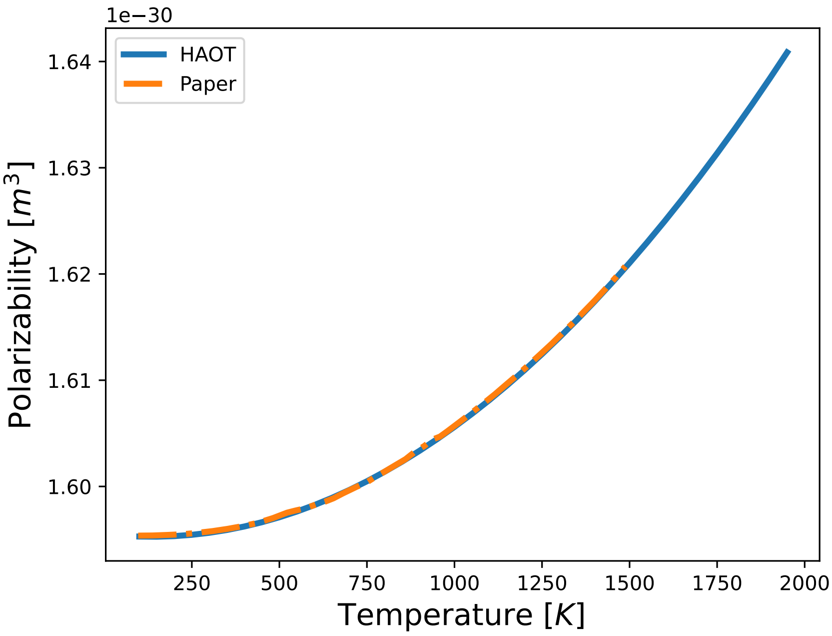


Figure 4.4: Polarizability of $O2$ for a wavelength of 633 [nm]

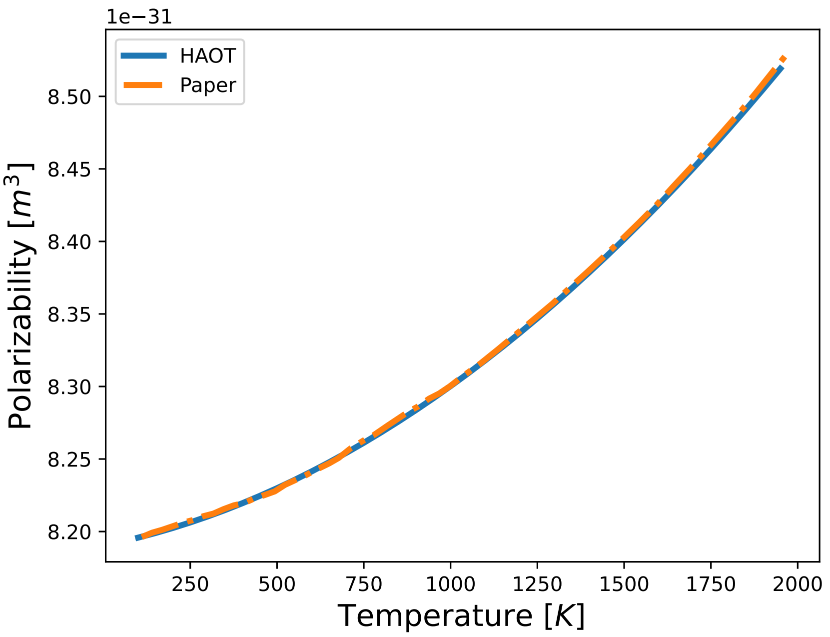


Figure 4.5: Polarizability of $H2$ for a wavelength of 633 [nm]

4.3 Quantum Mechanics

Figures 4.6 to 4.8 show the Boltzmann Distribution eq. (2.31), which was implemented using HAOT. As expected, the Boltzmann Distribution demonstrates that most of the energy is concentrated in the lower energy levels. Additionally, the temperature has an effect on the kurtosis; as the temperature increases, the kurtosis decreases.

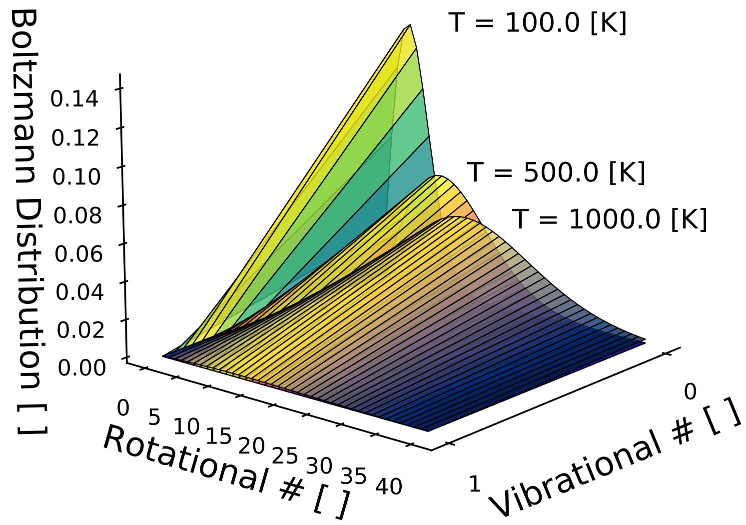


Figure 4.6: Harmonic Boltzmann Distribution for $N2$

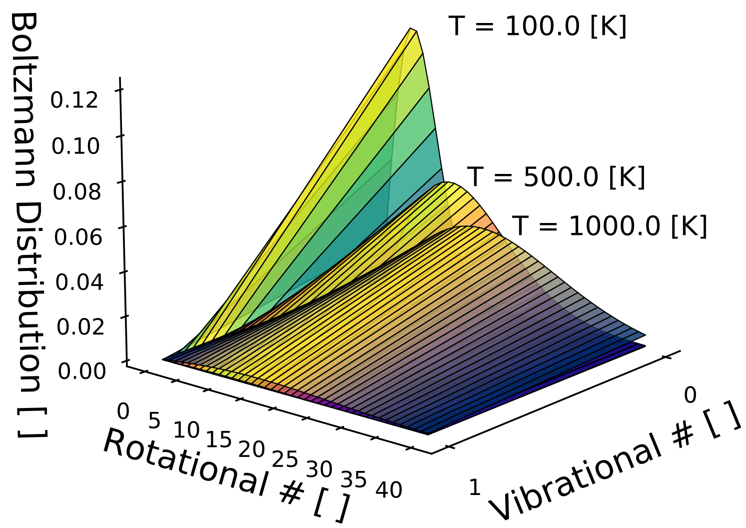


Figure 4.7: Harmonic Boltzmann Distribution for $O2$

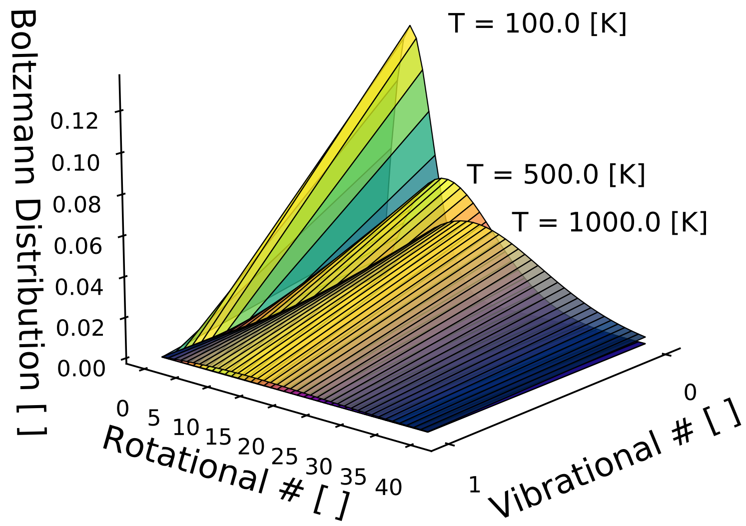


Figure 4.8: Harmonic Boltzmann Distribution for NO

Chapter 5

Numerical Results

This chapter presents the numerical results, obtained using the post-processing tool introduced in chapter 4. All the results in this chapter will use the same polarizability constants.

The equilibrium polarizability constants for neutral and ion species are provided on tab. 5.1 and tab. 5.2.

Species	$\alpha \times 10^{-30}$ $[m^3]$	Ref.
N	1.100	[93]
O	0.802	[93]
NO	1.700	[93]
N_2	1.740	[93]
O_2	1.569	[93]

Table 5.1: Equilibrium polarizability constants for neutral species

Species	$\alpha \times 10^{-30}$ $[m^3]$	Ref.
N^+	0.559	[94]
O^+	0.345	[95]
NO^+	1.021	[96]
N_2^+	2.386	[97]
O_2^+	0.238	[95]

Table 5.2: Equilibrium polarizability constants for ion species

5.1 Chemistry Composition

The purpose of this investigation is to study the nonequilibrium effects of aero-optics using a heat bath simulation. A heat bath simulation is an excellent tool for examining the time it takes for a gas to reach equilibrium after passing through a shock wave. The inputs for a heat bath simulation typically include the flow characteristics observed after a shock wave. These conditions are relatively straightforward to calculate in an experimental setup, facilitating a more direct comparison between computational and experimental results.

The solver used for this investigation is SU2 [98]. Tab. 5.3, shows the flow conditions for cases studied during this analysis. Most of these conditions were took from Hypersonic NonEquilibrium Comparison Cases (HyNECC) [99].

Case	$T_{tr}\,[K]$	$T_{rot}\,[K]$	$T_{vib}\,[K]$	$P\,[kPa]$
$1C$[99]	20000	20000	300	2760
$2C$[99]	10000	10000	300	1380
$3C$[99]	15000	15000	300	2067

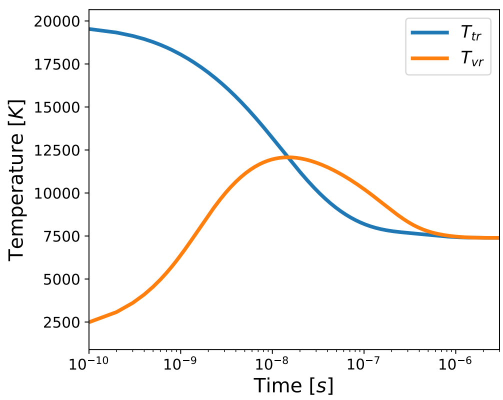
Table 5.3: Chemistry composition flow conditions

Tab. 5.4 shows the mass fraction composition for the cases in tab. 5.3.

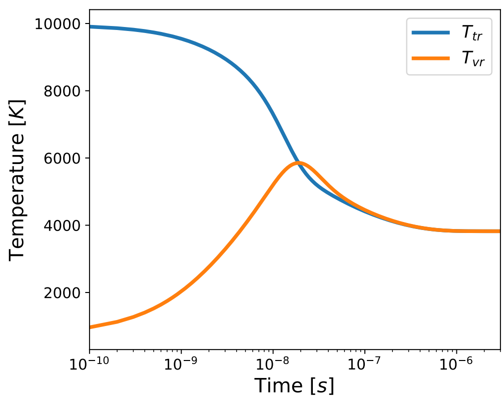
Case	N []	O []	NO []	N_2 []	O_2 []
$1C$	0.0	0.0	0.0	1.0	0.0
$2C$	0.0	0.0	0.0	0.0	1.0
$3C$	$6.5E-7$	0.2283	0.01026	0.75	$0.713E-3$

Table 5.4: Mass fraction composition for cases in tab. 5.3

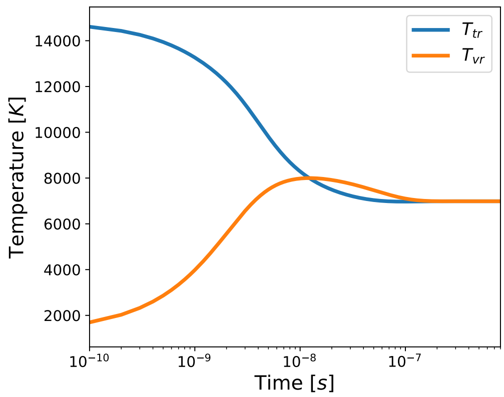
Fig. 5.1 illustrates the translational and vibrational temperatures for cases described in tab. 5.3 and tab. 5.4. As expected, the relaxation process takes longer for case $1C$, as it has the case with the highest initial translational temperature. In fig. 5.1, all cases reach thermal equilibrium at approximately 1 $[\mu s]$. The diatomic nitrogen case illustrated in Fig. 5.1(a) displays a higher energy transfer between the translational and vibrational modes.



(a) Case $1C$



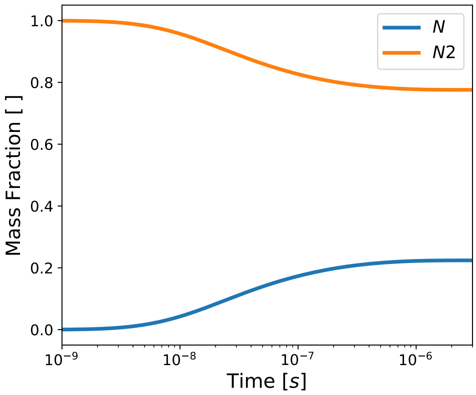
(b) Case $2C$



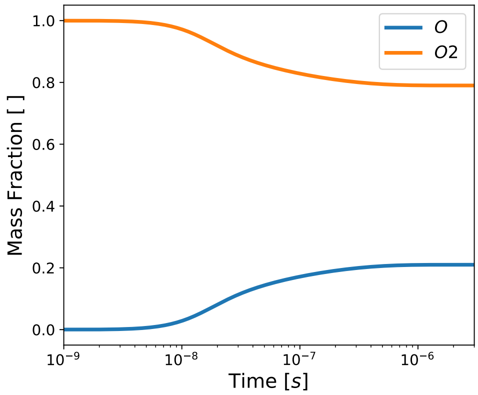
(c) Case $3C$

Figure 5.1: Temperature results for cases in tab. 5.3

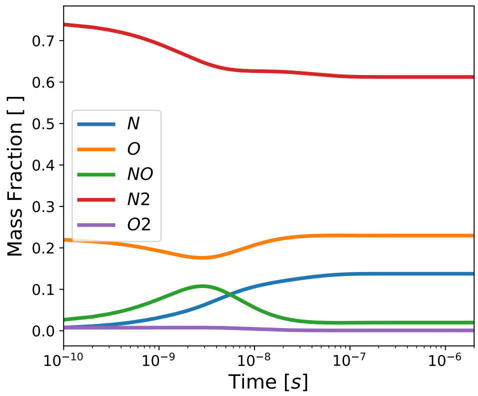
Fig. 5.2 shows the mass fraction composition for the cases described in tab. 5.3 and tab. 5.4. Fig. 5.2(a) shows the mass fraction for Case $1C$, which corresponds to a diatomic nitrogen gas. The diatomic nitrogen decomposes into nitrogen, and by the time (≈ 10 $[\mu s]$) the simulation reaches chemical equilibrium, approximately 20% of the diatomic nitrogen has decomposed into nitrogen. Fig. 5.2(b) shows the mass fraction for Case $2C$, which corresponds to diatomic oxygen gas.The diatomic oxygen decomposes into oxygen, and by the time (≈ 10 $[\mu s]$) the simulation reaches chemical equilibrium, approximately 20% of the diatomic oxygen has decomposed into oxygen. fig. 5.2(c) shows the mass fraction for Case $3C$, which corresponds to a 5 species air mixture. Some of the diatomic nitrogen and oxygen decomposed to a nitric oxide, with a peak value of approximately 10% at 5 $[ns]$.



(a) Case 1C



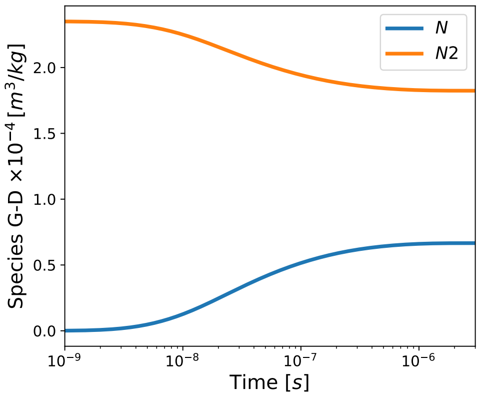
(b) Case 2C



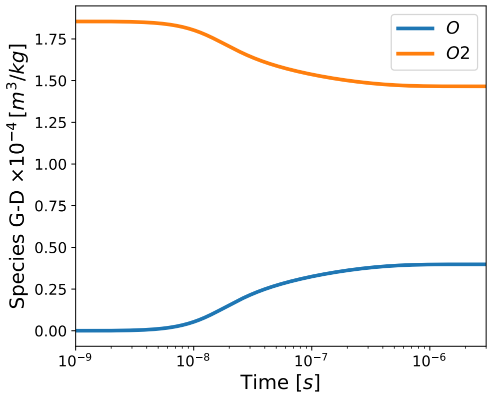
(c) Case 3C

Figure 5.2: Mass fraction results for cases in tab. 5.3

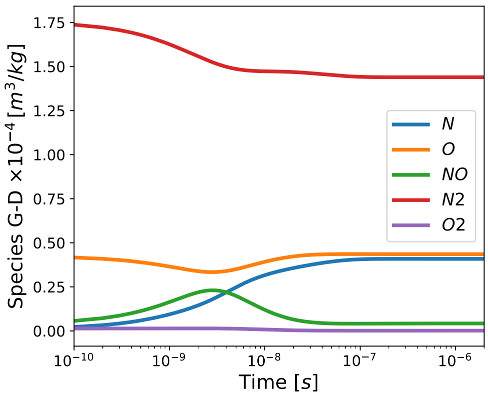
Fig. 5.3 shows the species G-D constants for the cases described in tab. 5.3 and tab. 5.4. These constants were calculated using eq. (2.18a). Fig. 5.3(a) shows the evolution of the diatomic nitrogen case. As expected, at the beginning of the simulation, the G-D constant for nitrogen is zero because the simulation contains only diatomic nitrogen. However, as the simulation evolves, some of the diatomic nitrogen gas dissociates into nitrogen causing the G-D constant for nitrogen to no longer be zero. Fig. 5.3(b) shows the evolution of the diatomic oxygen case. As expected, at the beginning of the simulation, the G-D constant for oxygen is zero because the simulation contains only diatomic oxygen. However, as the simulation evolves, some of the diatomic oxygen gas dissociates into oxygen causing the G-D constant for oxygen to no longer be zero. Fig. 5.3(c) illustrates the evolution of five gas species. Some of the diatomic nitrogen and oxygen recombine into nitric oxide, resulting in a peak in the G-D for nitric oxide at approximately 5 $[ns]$. However, this peak is short-lived as the nitric oxide quickly dissociates back into nitrogen and oxygen. This figure highlights the direct impact of density changes on the G-D. As is well known, in a nonequilibrium flow, the high temperatures cause molecular dissociation.



(a) Case 1C



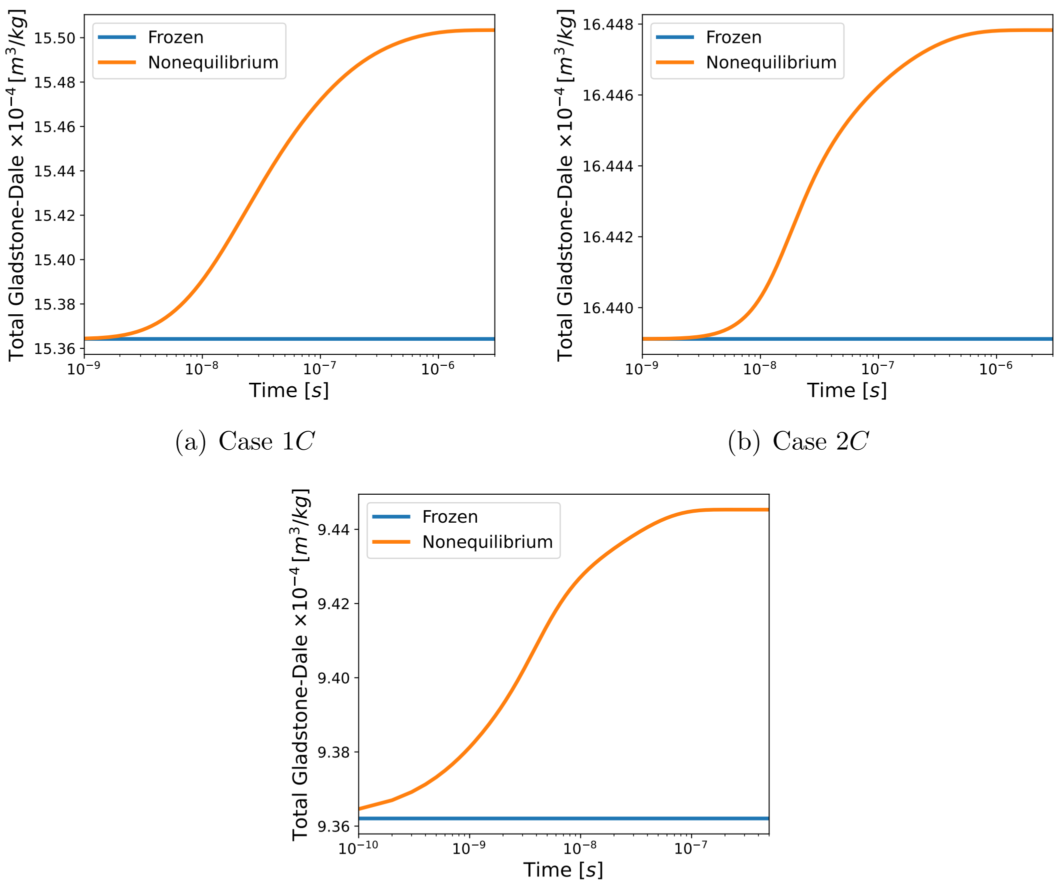
(b) Case 2C



(c) Case 3C

Figure 5.3: Species G-D results for cases in tab. 5.3

Fig. 5.4 compares the total G-D between a nonequilibrium case and a frozen flow case for the conditions outlined in tab. 5.4. These results indicate that a higher percentage of diatomic nitrogen corresponds to an increased G-D. This outcome aligns with expectations, as diatomic nitrogen exhibits the highest equilibrium polarizability among the species considered. Additionally, it is noteworthy that nonequilibrium effects contribute to an increase in the G-D.



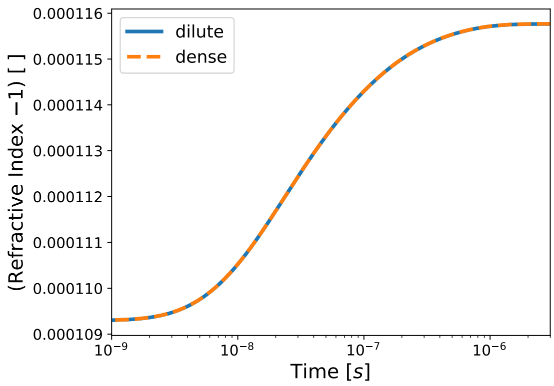
(a) Case 1C

(b) Case 2C

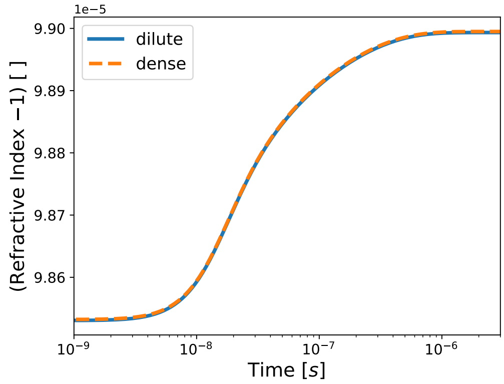
(c) Case 3C

Figure 5.4: Total Gladstone-Dale results for cases in tab. 5.3

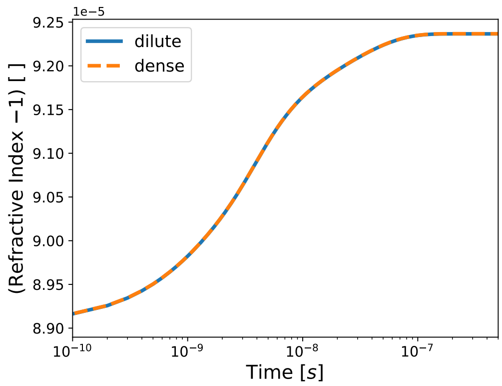
Fig. 5.5 presents the index of refraction for the cases described in tab. 5.3. The figure compares the dilute formulation (eq. (2.17a)) with the dense formulation (eq. (2.17b)) and tab. 5.4. It is observed that there is no significant difference between the dense and dilute formulations. As expected, the case shown in fig. 5.5(a), which contains a higher proportion of diatomic nitrogen, exhibits a greater change in the refractive index.



(a) Case $1C$



(b) Case $2C$



(c) Case $3C$

Figure 5.5: Index of Refraction results for cases in tab. 5.3

5.2 Species Study

This investigation compares the index of refraction between a 5 and 11 species gas. The solver used for this case was SU2 NEMO [100], which is the nonequilibrium version of SU2 that is coupled with MUlticomponent Thermodynamic And Transport properties for IONized gases in C++ (Mutation++) [101] library. The Park's $2T$ nonequilibrium model [87] model is used for this investigation.

Case	Species	T_{tr} $[K]$	T_{rot} $[K]$	T_{vib} $[K]$	P $[kPa]$
$1S1$[99]	5	15000	15000	300	2067
$2S1$	5	8000	8000	300	1000
$1S2$	11	15000	15000	300	2067
$2S2$	11	8000	8000	300	1000

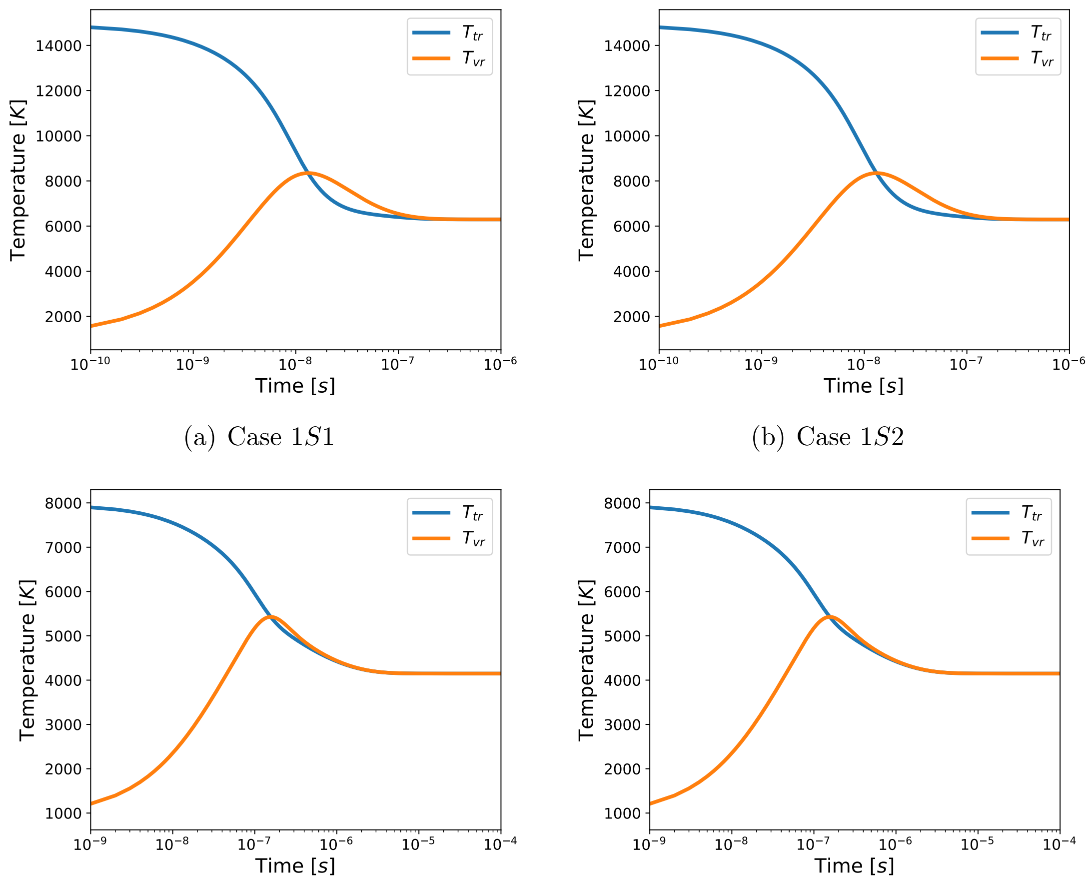
Table 5.5: Species studied cases

Cases in tab. 5.5 used the mass fraction composition for standard Air provided in tab. 5.6.

N []	O []	NO []	N_2 []	O_2 []
0.0	0.0	0.0	0.79	0.21

Table 5.6: Mass fraction composition for cases in tab. 5.5

Fig. 5.6 illustrates the translational and vibrational temperatures for the cases described in tab. 5.5 and tab. 5.6. The temperature plots compare a five-species gas (figures 5.6(a) and 5.6(c)) with an eleven-species gas (figures 5.6(b) and 5.10(d)). In the highest-temperature cases (figures 5.6(a) and 5.6(b)), thermal equilibrium is reached at approximately 0.5 $[ns]$. Notably, the temperatures do not appear to differ significantly between the five-species and eleven-species gases. Similarly, in the lowest-temperature cases (figures 5.6(c) and 5.10(d)), thermal equilibrium is achieved at approximately 5 $[ns]$, and again, there is no observable difference between the temperatures of the five-species and eleven-species gases.



(a) Case $1S1$

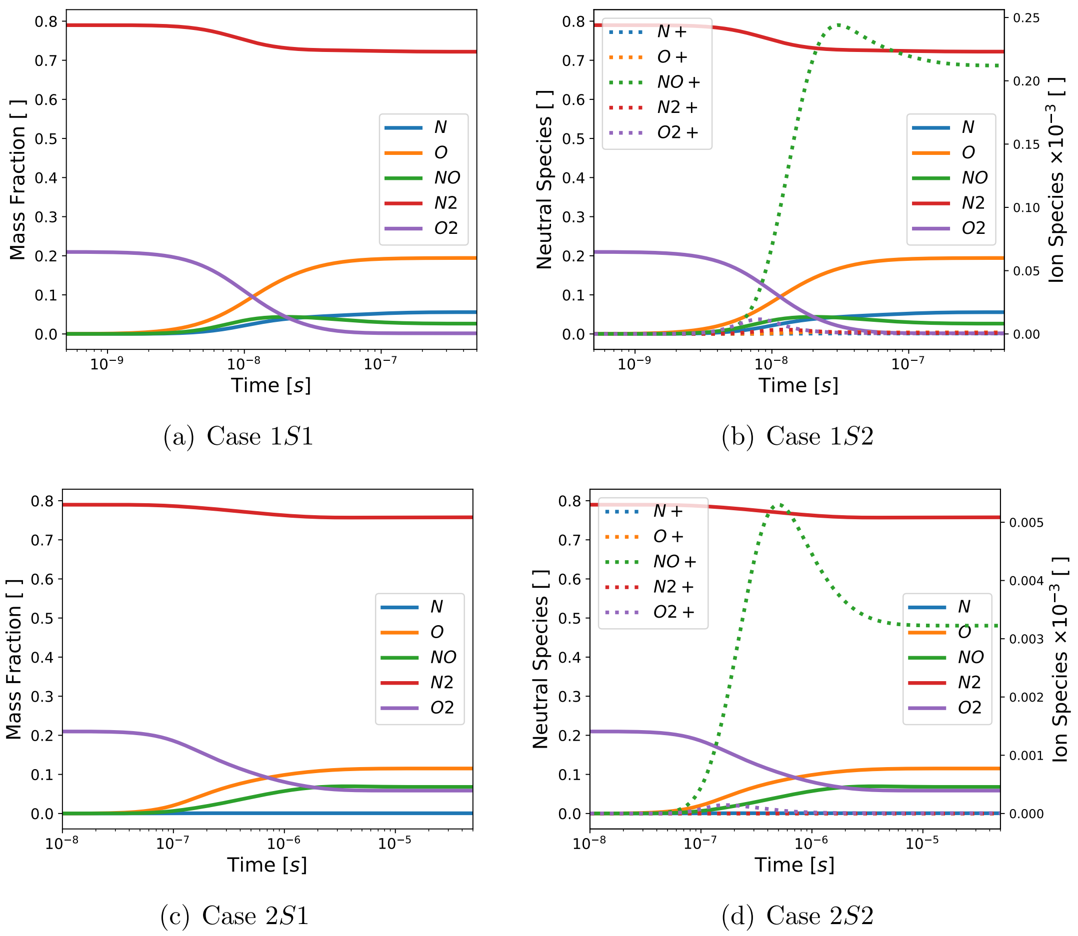
(b) Case $1S2$

(c) Case $2S1$

(d) Case $2S2$

Figure 5.6: Temperature results for cases in tab. 5.5

Fig. 5.7 illustrates the mass fractions for the cases described in tab. 5.5 and tab. 5.6. In the highest-temperature cases ($1S1$ and $1S2$), a higher rate of dissociation is observed, which is expected due to the greater energy available in the system. For the eleven-species case at the highest temperature (fig. 5.7(b)), a significant amount of diatomic nitrogen ions is generated, peaking at approximately 50 $[ns]$. However, the mass fraction of other ion species do not change significantly. In contrast, the lowest-temperature cases ($2S1$ and $2S2$) exhibit a lower rate of dissociation compared to the higher-temperature cases. In these cases, the maximum concentration of diatomic nitrogen ions occurs at approximately 50 $[ns]$. Although diatomic nitrogen ions are formed, their quantities remain negligible compared to the neutral species.



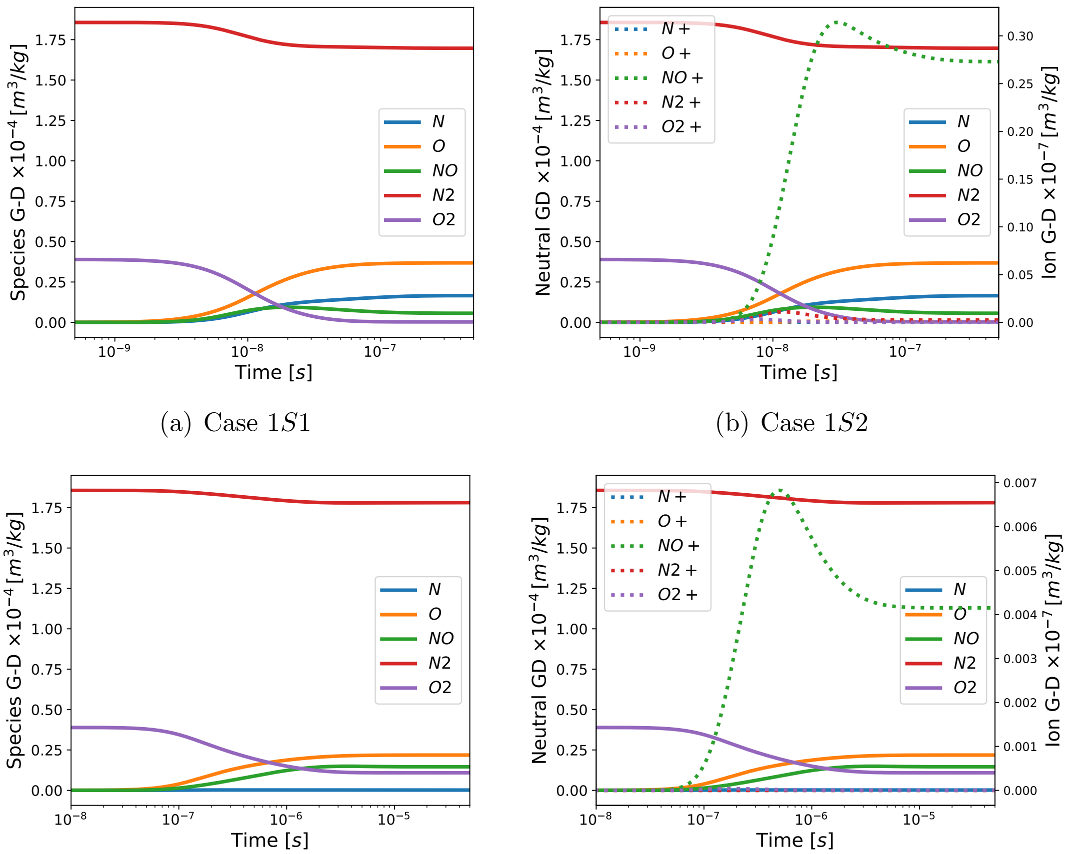
(a) Case 1S1 (b) Case 1S2

(c) Case 2S1 (d) Case 2S2

Figure 5.7: Mass fraction results for cases in tab. 5.5

Fig. 5.8 shows the species G-D for the cases described in tab. 5.5 and tab. 5.6. The eleven-species cases (1S2 and 2S2) demonstrate that the ionic part of the G-D constants are primarily influenced by the formation of charged nitric oxide. However, these values still appear to be low compared to those of the neutral species.

Comparing the five-species and eleven-species cases, it is evident that the ion contribution to the G-D is small—approximately one order of magnitude less for the highest-temperature case and two orders of magnitude less for the lowest-temperature case. Consequently, the ion contribution does not significantly impact the G-D.



(a) Case $1S1$

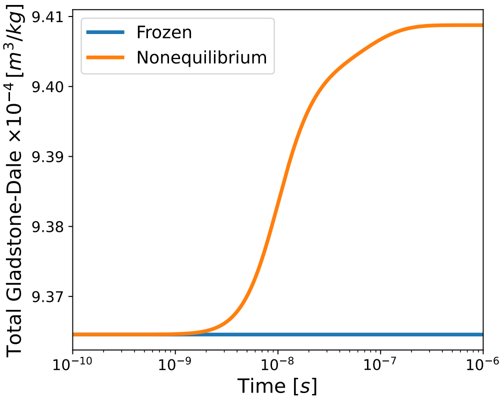
(b) Case $1S2$

(c) Case $2S1$

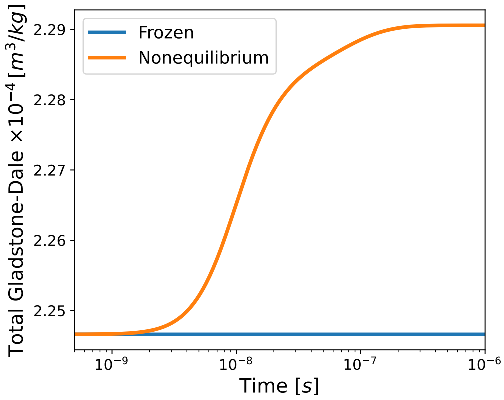
(d) Case $2S2$

Figure 5.8: Species G-D results for cases in tab. 5.5

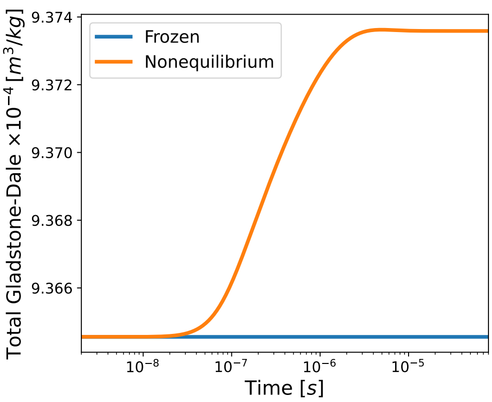
Fig. 5.9 shows the total G-D for the cases described in tab. 5.5 and tab. 5.6. As expected from fig. 5.8, the contribution from the ion species is minimal and not noticeable in the total G-D.



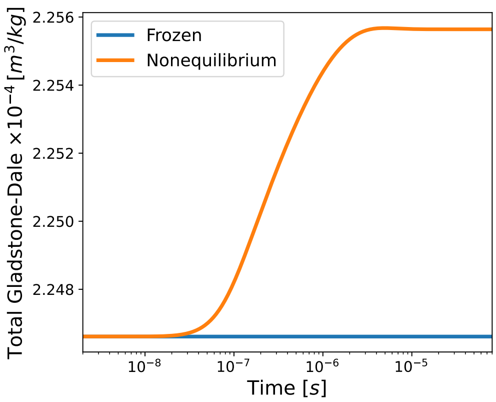
(a) Case $1S1$



(b) Case $1S2$



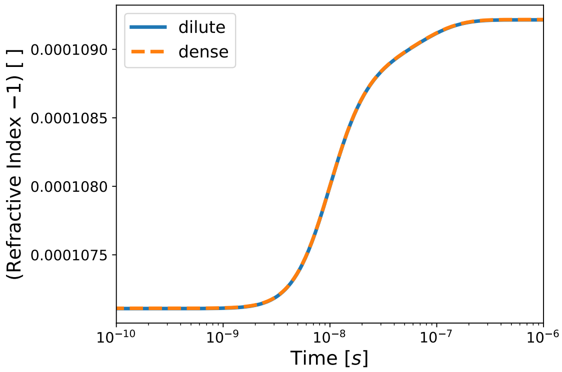
(c) Case $2S1$



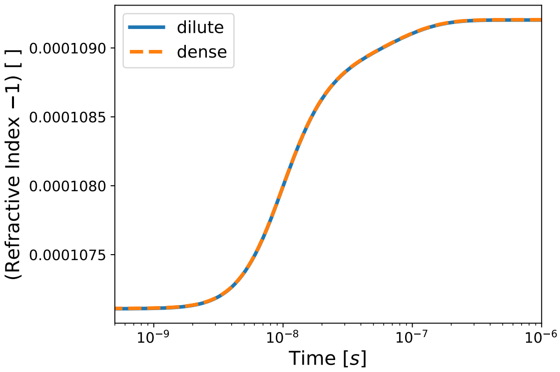
(d) Case $2S2$

Figure 5.9: Total G-D results for cases in tab. 5.5

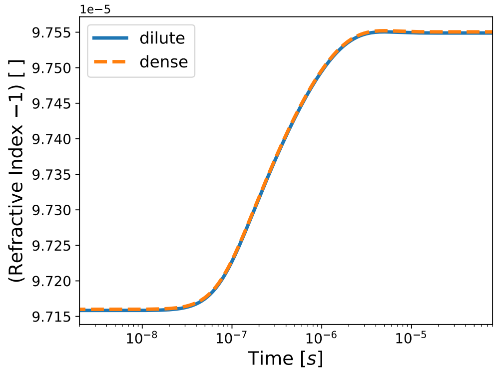
Fig. 5.10 shows the total index of refraction for the cases described in tab. 5.5 and tab. 5.6. As expected from fig. 5.10, the contribution from the ion species is minimal and not noticeable in the index of refraction. Hence, it can be concluded that using a five-species gas is sufficient to predict the index of refraction, making the computational cost less expensive.



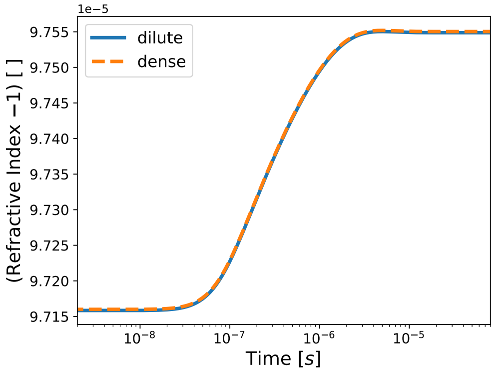
(a) Case $1S1$



(b) Case $1S2$



(c) Case $2S1$



(d) Case $2S2$

Figure 5.10: Index of Refraction results for cases in tab. 5.5

5.3 Chemistry Reaction Study

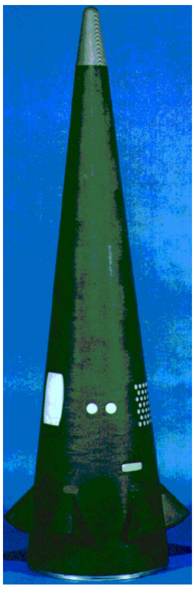
Use Results from AIAAA, laminar comparison between frozen and nonequilibrium flow

This section investigates nonequilibrium effects of aero-optics in a laminar flow by comparing a flow with nonequilibrium chemistry and the other one with frozen chemistry (chemical composition of the gas remains unchanged because the reactions are too slow to occur within the times scale of the flow). tab. 5.7 shows the cases used for this investigation.

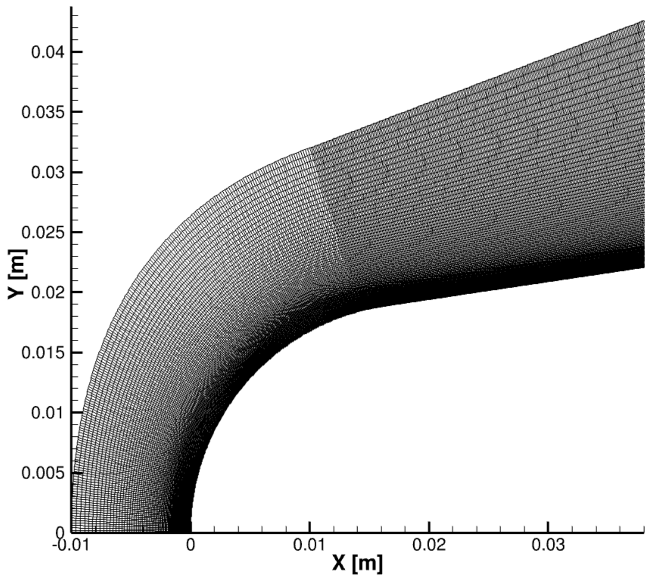
Case	M [ ]	T_{wall} $[K]$	T_∞ $[K]$	P_∞ $[Pa]$	h_0 $[MJ]$	q $[kPa]$
$1R$[28]	11.2	1000	293	1200	7.38	105
$2R$[28]	13.2	1000	293	1200	10.25	146
$3R$[28]	15.2	1000	293	1200	13.60	193

Table 5.7: Chemistry Study

For this investigation the IRV-2 vehicle, shown in fig. 5.11(a) is used. The IRV-2 vehicle is a sphere-biconic with a nose radius of 1.95 $[cm]$ and a total length of 1.386 $[m]$. Six drag flaps are located at the aft end of the vehicle and can be deployed to a maximum angle of 45 degrees [102]. For this work, only axisymmetric calculations on the IRV-2 nosetip are performed. fig. 5.11(b) shows the mesh used in this work.

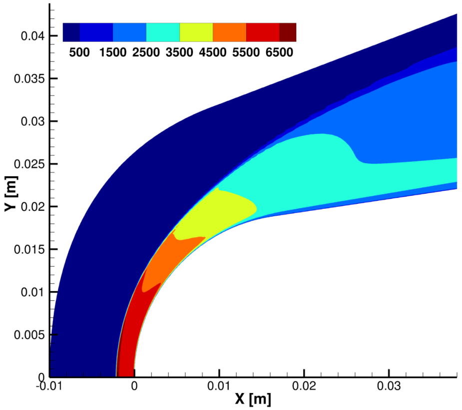


(a) IRV-2 vehicle

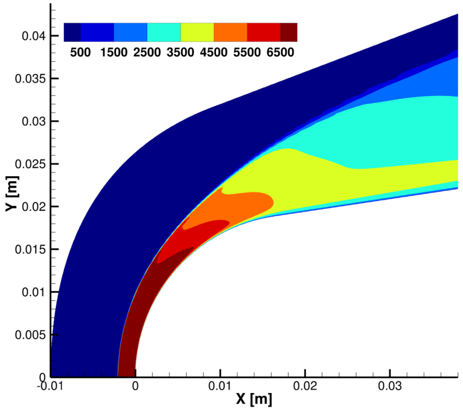


(b) IRV-2 mesh

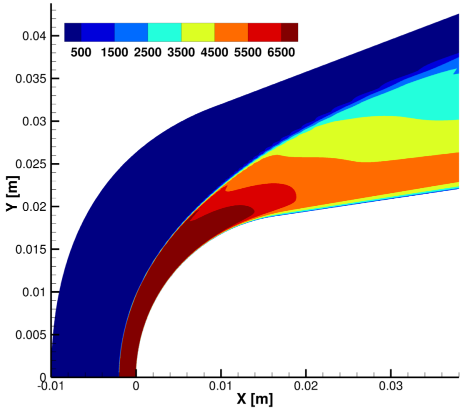
Figure 5.11: IRV-2



(a) Case $1R$

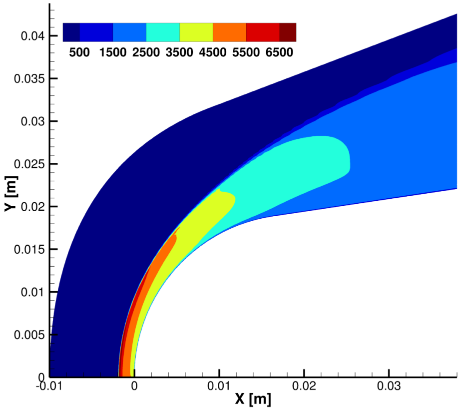


(b) Case $2R$

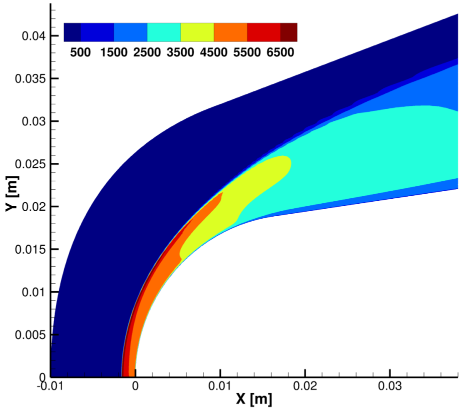


(c) Case $3R$

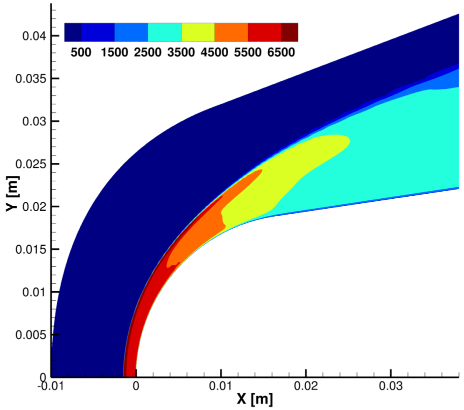
Figure 5.12: Contour translational temperature for frozen cases from tab. 5.7



(a) Case $1R$

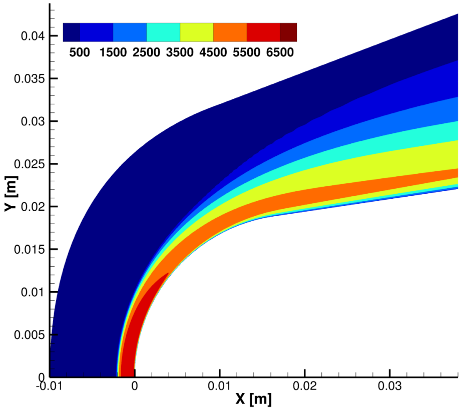


(b) Case $2R$

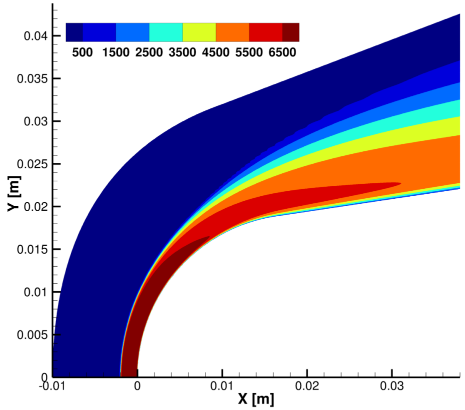


(c) Case $3R$

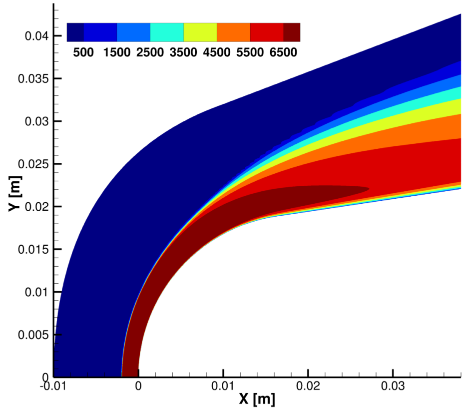
Figure 5.13: Contour translational temperature for nonequilibrium cases from tab. 5.7



(a) Case $1R$

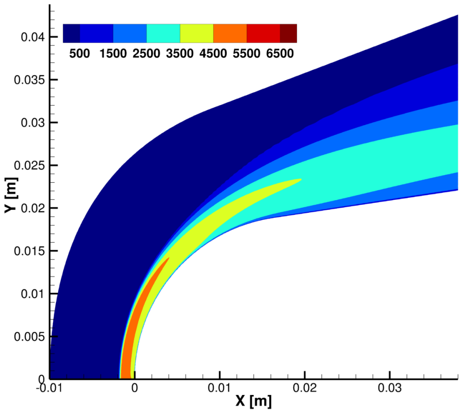


(b) Case $2R$

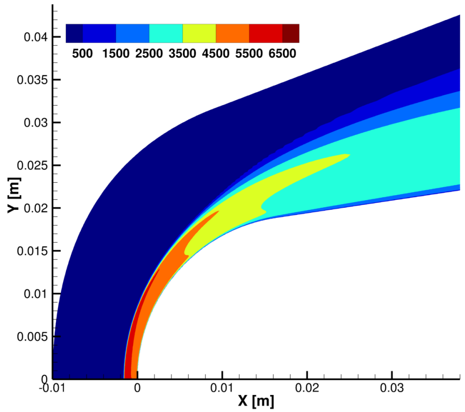


(c) Case $3R$

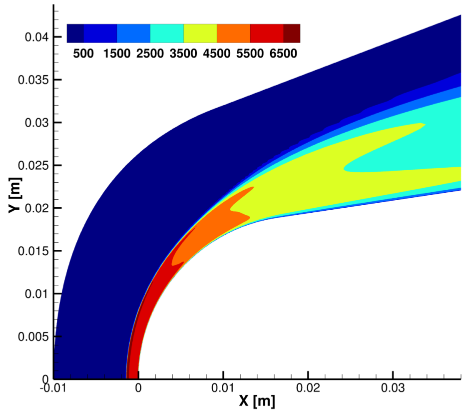
Figure 5.14: Contour vibrational temperature for frozen cases from tab. 5.7



(a) Case $1R$



(b) Case $2R$



(c) Case $3R$

Figure 5.15: Contour vibrational temperature for nonequilibrium cases from tab. 5.7

Figures 5.13(a) to 5.13(c) and 5.15(a) to 5.15(c) show temperature contour plots for cases described in table tab. 5.7. As expected the frozen flow cases over predict the translational and vibrational temperatures.

Add a why to what frozen flow overpredicts

5.4 RANS Study

Add the cases from the AIAA Paper

Case	M [ ]	T_{wall} $[K]$	T_∞ $[K]$	P_∞ $[Pa]$	h_0 $[MJ]$	q $[kPa]$
$1TR$[28]	21.17	300	219.47	3519.76		

Table 5.8: RANS Study

5.5 LES Study

LES investigationr

5.6 Ray Tracer Study

Chapter 6

Conclusion

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Appendices

Appendix A

Buldakov Expansion

$$E_{v,J} = \sum_{i,j=0} Y_{i,j} \left(v + \frac{1}{2}\right)^i \left[J\left(J+1\right)\right]^j$$

$$x = \left(\frac{r - r_e}{r_e}\right)$$

$$\alpha(r,\omega) = \alpha_e + \alpha_e' x + \frac{1}{2}\alpha_e'' x^2 + \frac{1}{6}\alpha_e''' x^3 \tag{A.1}$$

$$\left\langle v,J\right| \alpha^{(i)}(r,\omega) \left|v,J\right\rangle = \int \psi^*_{v,J}(r)\alpha_{r,\omega}\psi_{v,J}(r)dr$$

$$= \alpha_e \left\langle v,J|v,J\right\rangle + \alpha_e' \left\langle v,J\right| x \left|v,J\right\rangle + \frac{1}{2}\alpha_e'' \left\langle v,J\right| x^2 \left|v,J\right\rangle + \frac{1}{6}\alpha_e''' \left\langle v,J\right| x^3 \left|v,J\right\rangle$$

$$\begin{aligned}\langle v,J|\,\alpha^{(i)}(r,\omega)\,|v,J\rangle = {} & \alpha_e + \frac{1}{2}\left(2v+1\right)\left(-3a_1\alpha_e' + \alpha_e''\right)\left(\frac{B_e}{\omega_e}\right) + 4J\left(J+1\right)\alpha_e'\left(\frac{B_e}{\omega_e}\right)^2 + \\ & \left\{\alpha_e'\left\{-\frac{3a_1^3}{8}\left[15\left(2v+1\right)^2+7\right] + \frac{a_1a_2}{4}\left[39\left(2v+1\right)^2+23\right] - \right.\right. \\ & \left.\frac{15a_3}{4}\left[\left(2v+1\right)^2+5\right]\right\} + \alpha_e''\left\{\frac{a_1^2}{8}\left[15\left(2v+1\right)^2+7\right] - \right. \\ & \left.\left.\frac{3a_2}{4}\left[\left(2v+1\right)^2+5\right]\right\} - \frac{a_1}{24}\alpha_e'''\left[15\left(2v+1\right)^2+7\right]\right\}\left(\frac{B_e}{\omega_e}\right)^2 + \\ & \left\{\alpha_e'\left[27a_1\left(1+a_1\right)+24\left(1-a_2\right)\right] - \right. \\ & \left.3\alpha_e''\left(1+3a_1\right) + \frac{\alpha_e'''}{8}\right\}J\left(J+1\right)\left(2v+1\right)\left(\frac{B_e}{\omega_e}\right)^3\end{aligned} \tag{A.2}$$

$$\begin{aligned}& \frac{1}{2}x^2\left(\frac{\omega_e}{2B_e}\right)^{2/2} + \frac{a_1}{2}\left(\frac{2B_e}{\omega_e}\right)^{1/2}x^3\left(\frac{\omega_e}{2B_e}\right)^{3/2} + \\ & \frac{a_2}{2}\left(\frac{2B_e}{\omega_e}\right)x^4\left(\frac{\omega_e}{2B_e}\right)^{4/2} + \frac{a_3}{2}\left(\frac{2B_e}{\omega_e}\right)^{3/2}x^5\left(\frac{\omega_e}{2B_e}\right)^{5/2} + \\ & \frac{a_4}{2}\left(\frac{2B_e}{\omega_e}\right)^2x^6\left(\frac{\omega_e}{2B_e}\right)^{6/2} + \frac{a_5}{2}\left(\frac{2B_e}{\omega_e}\right)^{5/2}x^7\left(\frac{\omega_e}{2B_e}\right)^{7/2} + \\ & \frac{a_6}{2}\left(\frac{2B_e}{\omega_e}\right)^3x^8\left(\frac{\omega_e}{2B_e}\right)^{8/2}\end{aligned} \tag{A.3}$$

$$\begin{aligned}& \frac{J\left(J+1\right)}{2}\left(\frac{2B_e}{\omega_e}\right) - J\left(J+1\right)\left(\frac{2B_e}{\omega_e}\right)^{3/2}x^1\left(\frac{\omega_e}{2B_e}\right)^{1/2} + \\ & \frac{3J\left(J+1\right)}{2}\left(\frac{2B_e}{\omega_e}\right)^2x^2\left(\frac{\omega_e}{2B_e}\right)^{2/2} - 2J\left(J+1\right)\left(\frac{2B_e}{\omega_e}\right)^{5/2}x^3\left(\frac{\omega_e}{2B_e}\right)^{3/2} + \\ & \frac{5J\left(J+1\right)}{2}\left(\frac{2B_e}{\omega_e}\right)^3x^4\left(\frac{\omega_e}{2B_e}\right)^{4/2}\end{aligned} \tag{A.4}$$